

More factors matter and factors matter more than you might think:

The role of time variation in factor premia*

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Abstract

Allowing for time variation in factor premia, we show that portfolios formed from factor principal components (PCs) yield economically large out-of-sample Sharpe ratios that increase on average until about forty PCs are employed. That is, time-varying factors have strong predictive power for the cross-section of stock returns, and to a substantial extent are not redundant of each other. The SDF implied by this time-varying approach displays smaller pricing errors as compared to canonical factor models not allowing for time variation. The central findings are robust to alternative methods of extracting PCs, to the construction and use of alternative factors formed from WRDS financial ratios, and to the exclusion of low volume or small stocks. These results imply that barriers to cross-factor arbitrage are large and indicate roles for economic complexity and investor learning in asset pricing.

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I. Introduction

The literature has identified hundreds of variables, including firm characteristics and “factors” constructed as returns to long-short portfolios, that appear to have significant ability to predict the cross-section of stock returns. However, more recent research has asserted significant progress in taming what the literature refers to as the “factor zoo.” Hou, Xue, and Zhang (2020) report that the first four principal components (PCs) of the 158 characteristics they study explain over half the sample variation, and that eight PCs can explain more than two thirds of the variation. Kelly, Pruitt, and Su (2019, henceforth KPS) use the method of instrumented principal components analysis (IPCA), and report that five latent IPCA factors explain the cross-section of returns better than alternative factor models. Kozak, Nagel, and Santosh (KNS, 2018) study Sharpe ratios (SRs) for portfolios formed from factor PCs, and report that “while factors beyond the first few PCs contribute substantially to the maximum SR in sample, PCs beyond the first few no longer add to the SR out of sample.” Collectively, this evidence suggests that the many characteristics and characteristic-based factors identified in the literature contain a substantial amount of redundant information, and that only a few sources of independent variation are required to explain the cross section of stock returns.¹

We show that allowing for time variation in factor return premia leads to notably different conclusions. In contrast to relatively opaque machine learning applications (e.g., Kelly, Malamud, and Zhou, 2024), we rely on a simple and transparent approach: rolling sixty-month in-sample estimation periods. We follow KNS and others in focusing on out-of-sample (OOS) Sharpe ratios, as these are informative both regarding the number of factors with distinct explanatory power for the stochastic discount factor (SDF) and the economic importance of factors’ forecast power for the cross-section of stock returns. We show that OOS Sharpe ratios for portfolios formed from factor PCs are economically

¹ It is well-known (e.g., Cochrane, 2005, Chapter 7) that any multi-factor model can be restated as a single factor representation that prices assets just as well. For example, the two factor model $m = a + bF_1 + cF_2$ (where b and c are non-zero constants) can be restated as the single factor model $m = a + bF^*$ where $F^* = F_1 + (c/b)F_2$. Yet, in this setting each of the two factors contains non-redundant information necessary to define F^* or to explain its variation. More broadly, it is a meaningful question to ask how many observable factors contribute information relevant to the formation of the single factor (or other low-dimensional) representation. This paper addresses this question.

large and increase on average as portfolios are formed from as many as about forty factor PCs. These conclusions hold in the same data studied by KNS (2018), and are even stronger in a broader set of 212 factors compiled by Chen and Zimmerman (2022, henceforth CZ), and made available by them at openassetpricing.com. Our results imply that empirically observable factors contain substantial information relevant to identifying the SDF and are useful in forecasting the cross-section of stock returns. Further, the fact that OOS Sharpe ratios continue to increase as portfolios are formed from larger numbers of PCs implies that the factors identified in the literature contain information that, to a substantial extent, is not redundant in linear combination. Remarkably, our central conclusions obtain even when portfolios are formed from factor PCs in the simplest possible method, namely by equal-weighting.

The reasoning that only a few sources of common variation should be necessary to explain the cross section of returns is well articulated by KNS (2018, page 1184): “the existence of a relatively small number of arbitrageurs should be sufficient to ensure that near-arbitrage opportunities—that is, trading strategies that earn extremely high Sharpe ratios (SRs)—do not exist.” Similar reasoning is expressed by Cochrane and Saa-Requejo (2000), who refer to “good deal” bounds that limit Sharpe ratios. KNS argue that arbitrage trading across factors should eliminate return premia not associated with high covariation in factor returns, and that, as a consequence, the stochastic discount factor “can be represented to a good approximation” with a “few dominant factors” that can be extracted via a small number of factor principal components.

In contrast to KNS’ conclusions, we document high OOS Sharpe ratios when allowing for time variation in factor premia, in the identical sample they studied. The results are even stronger in the broader sample of 212 factors compiled by Chen and Zimmerman (CZ, 2022). As robustness tests, we show that our central results are also obtained in factors that are constructed based on the WRDS Financial Ratios and in latent factors extracted from firm characteristics using the IPCA method of KPS (2019). The evidence that allowing for time variation leads to higher OOS Sharpe ratios is consistent with the implications of conditional asset pricing models where parameters vary through time, but not so

abruptly that estimates formed during relatively short in-sample periods become irrelevant in the subsequent OOS periods.

We also show that the return premia associated with individual factors typically display multiple alternating periods of significance and insignificance, both within and outside of the time periods studied by the original authors, to an extent that cannot be attributed to random variation around constant factor premium. We document that the numbers of factors with significant explanatory power for the cross-section of returns at various points in time are related to diversity in firm characteristics, macroeconomic conditions, and measures of economic complexity. These findings challenge the perspective that factor return premia in a complex and dynamic economy necessarily permanently decline in magnitude and/or disappear after being documented.

Our central finding, that allowing for time variation in factor premia leads to the implication that more than a few factor PCs are relevant in explaining the cross section of stock returns, is robust. The findings persist even when (1) we extract factor PCs by methods that consider the mean as well as the variance of factor returns, as in Lettau and Pelger (2020) or Bryzgalova et al. (2023), (2) we reconstruct factors to exclude small capitalization or low turnover stocks, (3) we rely on factors constructed using the WRDS financial ratios instead of factors discovered by prior researchers, and (4) we employ the IPCA method of Kelly et al. (2019) to construct latent factors that incorporate time variation in factor betas. We also show that allowing for reasonable estimates of trading and shorting costs reduces, but does not eliminate, the excess of the OOS Sharpe ratios we document compared to the overall market SR during our sample period. Further, we show that the SDF estimated by our simple time varying approach leads to improved asset pricing performance when compared to canonical models that are implemented without allowing for time variation in parameters, including the Fama and French (1993) three factor model, the Fama and French (2015) five factor model, the Hou, Xue, and Zhang (2015) Q factor model, as well as the five PC approach of KNS (2018).

Our results need not imply “money left on the table,” but rather suggest that cross-factor arbitrage activity in a dynamic economy faces larger barriers than have been previously recognized. This outcome

may reflect in part that arbitrage capital is slow moving, as described by Mitchell, Pedersen, and Pulvino (2007). The need for investors to learn about or estimate the current state of time-varying factor premia is likely to comprise an additional barrier to arbitrage, that is likely to be greater at times of heightened economic complexity. In addition, frictions can imply that the marginal investor differs across the stocks that comprise the long and short legs of various factors, along the lines of the reasoning and evidence presented by Dew-Becker and Giglio (2023). For example, Kojen and Yogo (2019) document that institutions hold only small sets of stocks that change slowly over time. Many institutions may be wholly restricted from trading in highly volatile or illiquid stocks. In our view, the evidence of high OOS Sharpe ratios for portfolios formed based on factor PCs implies that barriers to cross-factor arbitrage are substantial, and that the determinants of expected stock returns are more complex than canonical models would imply. It is also noteworthy that returns are related to observable factors, not only latent factors (as documented by Kelly, Pruitt and Su, 2019). Observable factors can be studied with the goal of identifying the economic mechanisms that give rise to time-varying return predictability.

A potential concern is that allowing for time variation in factor premia may provide additional scope for collective specification searches or other sources of bias. We avoid amplifying any such bias in this study by employing the relatively standard 60-month horizon for our rolling estimation. Further, we examine the relation between factor premia and measures of firm and economic complexity that were not directly employed to identify the factors, showing that time variation in the number of economically relevant factors is significantly related to measures of the macroeconomic environment, including the number of firms that are listed, the cross-firm diversity of characteristics, and an economic complexity index. On balance, the results presented here indicate significant roles for time-varying economic complexity and for barriers to cross-stock arbitrage in asset pricing.

II. Placing our Study in the Literature

We are certainly not the first to consider potential time variation in factor premia. However, the focus of prior studies has mainly been on canonical factors such as the overall market, firm size or value

(market-to-book ratio).² We believe it is desirable to examine larger sets of factors in order to assess whether a narrow or broad set of factors can contribute independent information useful to forecasting the cross-section of stock returns, and as a window into the effectiveness of the cross-factor arbitrage that is central to models of empirical asset pricing. Haddad, Kozak, and Santosh (2020) and Kelly, Malamud and Pedersen (2023) take steps in this direction when they show that time variation in the return premia associated with certain factors can be estimated based on the portfolios' market-to-book ratios. We relax the restriction that factor premia are functions of market-to-book ratios alone and study a broader and time-varying set of factors. Having done so, we show that time-varying factor return premia do not exclusively decline in magnitude or disappear after publication of the original evidence.

While we refer throughout this paper to time variation in factor premia, the economic relevance of factors can vary due to changes in either a factor's return premium per unit of risk or the quantity of such risk (i.e., volatility).³ **We do not seek to decompose the variation into these components.** We study t -statistics estimated for factor alphas (which are proportional to factor-level information ratios), which reflect both mean returns and volatility. OOS Sharpe ratios also capture both aspects. We do not take a stand as to whether the return premia arise because of investor aversion to undesirable factor outcomes, mispricing in the face of barriers to arbitrage, or a combination thereof.

The evidence of economically substantive time variation in factor premia is related to the literature on investor learning. Lewellen and Shanken (2002) note that rational investors' uncertainty regarding parameters such as the dividend growth rate allow for return predictability to arise in equilibrium. Martin and Nagel (2022) show how investors' incorrect priors regarding parameters of the

² See, for example, Conrad and Kaul (1988), Ferson and Harvey (1991), Cochrane (1999), van Dijk (2011) and Ehsani and Linnainmaa (2021). The literature has not always been consistent in usage of the terms "characteristic" and "factor." To be precise, we use the term "characteristic" to refer to a firm-level attribute, such as firm size or profitability, and we use the word "factor" to refer to returns on a long-short portfolio. More specifically, each factor is the time series of returns on a portfolio that is long a set of stocks selected with either left or right tail outcomes on a given characteristic, e.g., firms of small size or high profitability, and short a set of stocks with outcomes in the opposite tail, e.g., stocks of large size or low profitability. We do not herein use the term factor to refer to outcomes obtained by the statistical method of factor analysis.

³ That return premia can vary over time relative to canonical factors' volatility is documented for example in Moreira and Muir (2017). We extend this insight to the hundreds of non-canonical factors.

return distribution in combination with complexity in the form of many relevant firm characteristics allows for out-of-sample return predictability as investors learn about the true distribution, even if the underlying economic structure is stable. Chinco, Neuhierl, and Weber (2021), also assuming constant underlying parameters, assess how traders optimally combine their prior assessments with the emergence of empirical evidence to determine which signals they will attempt to trade on. We reason that a dynamic economic environment with time-varying parameters only heightens the importance of investor learning as a barrier to arbitrage, thereby providing scope for factor premia that are economically large at times and modest at other times. Machine learning approaches may allow more precise forecasts of factor premia, particularly if these methods are extended to allow for time variation in relevant parameters.⁴ We contribute by documenting significant effects of time variation in factor risk premia using relatively simple and transparent methods.

Our findings that time variation in factor premia is economically important highlight the relevance of the considerable body of evidence regarding investor and firm diversity. Investors differ in terms of both their sophistication and their investment objectives, which allows that the identity of the marginal investor can differ across stocks and, in each stock, can change over time.⁵ Further, some investors hold positions for extended periods, others periodically rebalance to target weights, and yet others trade episodically in response to wealth shocks, opportunities to provide liquidity or correct perceived mispricings, or in response to rumors. Some investors trade directly, while others delegate portfolio decisions to professional managers, whose objectives can differ from those of their investors due to agency issues arising, for example, from specific compensation plans (e.g., Kashyap, Kovrijnykh, Li and Pavlova, 2021) or as a function of career horizons. The trades of professional investors can depend on considerations such as the funding liquidity of their employing firms, and return premia have also been

⁴ See, for example, Gu, Kelly, and Xiu (2020) and Gu, Kelly and Xiu (2021), Giglio, Kelly and Xiu (2022), and Kelly, Malamud, and Zhou (2024).

⁵ Recent studies documenting the diversity of individual trading approaches include Barber, Huang, Odean, and Schwarz (2022), Chen, Kumar, and Zhang (2021) and Bali, Brown, Murray Tang (2017).

shown to depend on the leverage of financial sector firms.⁶ Consistent with the reasoning that return premia can depend on the identity of the marginal investor, Betermier, Calvet, Knüpfer, and Kvaerner (2021) show that the cross-section of expected stock returns depends in part on the proportion of individual investors that are younger as well as the proportion that are wealthier. We shed light on the question of whether arbitrageurs are able to eliminate any inconsistencies in pricing that may arise because the marginal investor potentially differs across firms and time.

Firms, like investors, also change over time. The economic characteristics of newly listed firms often differs from those of existing firms, as shown by Campbell, et al. (2001), Fama and French (2004), and Kahle and Stulz (2017) and the characteristics of existing firms evolve. We construct a measure of cross-sectional diversity in the observable characteristics that are collectively known to be related to expected returns and show the number of factors that are significant in explaining stock returns increases with such diversity.

We present evidence that the trading cost estimates employed by Kelly et al. (2023) reduce but do not eliminate the high OOS Sharpe ratios implied by our time varying approach. It is possible that a deeper investigation might support the conclusion that, in line with Lewellen (2011) and Chen and Velikov (2023) who study narrower samples, no excess returns could be captured. However, we believe it is important to understand the nature of cross-sectional variation in expected returns and time series variation therein, whether such trading opportunities exist or not. As one example, researchers are interested in knowing if illiquidity affects expected returns, as implied by Amihud and Mendelson (1986), among others, even if illiquidity costs themselves prevent profit opportunities to an active trader. Further, our results imply time variation in firms' costs of capital, even if they do not imply arbitrage profits. In any case, our evidence is useful both in assessing the number of factors that provide distinct information relevant to the cross-section of expected stock returns and in terms of information regarding the effective limits to cross-factor arbitrage.

⁶ See, for example, He, Kelly and Manela (2017), Adrian, Etula, and Muir (2014) and He and Krishnamurthy (2013).

III. Time Variation in Factor Premia and the Cross Section of Stock Returns

In this paper we mainly study 212 factors drawn from Chen and Zimmerman (2022). However, before reporting results based on this broad sample, we demonstrate that time variation in factor premia is empirically important even in the narrower and more familiar samples employed by Kozak, Nagel and Santosh (KNS, 2018). They study a sample comprised of the long and short legs of fifteen anomaly-based factors from Novy-Marx and Velikov (2016), as well as a sample comprised of twenty-five size and book-to-market based portfolios formed in the manner of Fama and French (1993).

a. Initial Evidence: Allowing for time variation in the KNS (2018) sample

Kozak, Nagel, and Santosh (2018) (KNS) constitutes a prominent example of recent studies that conclude there are only a few independent sources of priced variation in the cross section of stock returns. They employ a split-sample approach that does not accommodate time variation in parameters; the first twenty-five years of their 1965 to 2015 sample period is used for in-sample estimation of factor principal components, and the second twenty-five years comprises the out-of-sample period. They report that as few as four factor PCs capture the relevant information, as out-of-sample (OOS) Sharpe ratios do not meaningfully increase when portfolios are formed from more than four factor PCs.⁷

We study the same data as KNS, while allowing for flexible time variation in parameters. Following KNS and others, we extract and study factor PCs that are identified in the standard manner by maximizing the sample variance explained by each PC in turn. An advantage, beyond dimension reduction, for using PCs to form efficient portfolios as compared to using the larger sample of anomaly portfolios directly is that the 'test assets' are nearly orthogonal, which improves the estimation of potentially unstable covariance-based weights. Of course, principal component analysis does not comprise the only potential method of reducing the dimensionality of the information contained in a large number of factors. We rely on standard PC analysis as our baseline for two reasons. First, this ensures

⁷ Kozak, Nagel, and Santosh (2020) provide complementary evidence. While they focus on OOS R^2 statistics rather than Sharpe ratios, they also report that only a few factor PCs are relevant. However, their analysis, like KNS (2018), is predicated on parameters being constant across time.

that our outcomes differ from prior studies that relied on PC analysis primarily (and in the case of the KNS data entirely) because of our allowance for time variation in factor return premia. Further, the PC method extracts information on the factor comovement that is central to KNS' arbitrage argument. The magnitudes of OOS Sharpe ratios based on factor PCs allow for the assessment of whether linear factor combinations contain redundant information and provide insights into the effectiveness of the cross-factor arbitrage envisioned by KNS and others. However, we also report outcomes when extracting factor PCs by the methods of Lettau and Pelger (2020) and Bryzgalova et al. (2023), which place weight on factor means as well as variances.

In Figure 1 we report in-sample and out-of-sample Sharpe ratios obtained for the KNS data. All Sharpe ratios we report are annualized. Panels A and B of Figure 1 pertain to portfolios constructed from the PCs of the long and short legs of fifteen anomaly-based factors, while Panels C and D report corresponding outcomes for portfolios formed from the PCs of twenty-five factors constructed based on firm size and book-to-market equity ratios. The red line displays outcomes obtained using the KNS split-sample approach and replicates their results: factor PCs beyond the first few contribute substantially to in-sample Sharpe ratios, but PCs beyond the first four do not substantially enhance OOS Sharpe ratios.

We compile results that parallel those reported by KNS in their Figure 4, while allowing for time variation in factor return premia by use of rolling 60-month estimation windows.⁸ Our baseline approach (applied here to the KNS data, and in subsequent sections to the CZ factors, factors constructed from the WRDS financial ratios and latent factors obtained by use of the KPS (2019) approach), is as follows. We extract PCs from factors within each rolling in-sample period $t-59$ to $t = 0$, using standard methods as in KNS, and refer to weights on factors in the PCs as factor loadings. We construct OOS PCs for months $t+1$ to $t+36$ by applying in-sample factor loadings to the corresponding monthly factor returns. We form investment portfolios from PCs, with weights chosen to maximize the in-sample Sharpe ratio, i.e., to

⁸ To maintain compatibility, we measure out-of-sample results from the same 25 years that comprise the out-of-sample period for Kozak, Nagel and Santosh (2018). That is, when employing N months to define the in-sample period, we begin the estimation with the final N months of the first half of the sample, so that the first out-of-sample month is identified starting at the sample midpoint.

identify the in-sample tangency portfolio.⁹ Our object of study is the OOS Sharpe ratio across months $t+1$ to $t+36$ for portfolios formed using in-sample weights and OOS PC returns.

The blue lines on each Panel of Figure 1 display Sharpe ratios obtained from the KNS data for portfolios formed from varying numbers of the resulting factor PCs. Two points are noteworthy. First, Sharpe ratios, both in- and out-of-sample, are virtually all larger when time variation in factor premia is accommodated than when it is not. The divergence in Sharpe ratios is apparent in each Panel of Figure 1 but is greater for the factors formed based on size and book-to-market. Second, Sharpe ratios increase as more factor PCs are employed. As KNS note, this is a mechanical outcome for in-sample Sharpe ratios. Importantly, this is not a mechanical result out-of-sample. That is, accommodating time variation in factor premia allows observation of the fact that factor PCs beyond the first few contain information that forecasts the cross-section of returns in the KNS sample, while suppressing time variation obscures the finding.

The increases in OOS Sharpe ratios when allowing for time variation are large. In particular, focusing on portfolios constructed based on the widely-studied firm size and book-to-market characteristics (Panel D), OOS Sharpe ratios formed from twenty to twenty-five factor PCs are more than twice as large when time variation is allowed for as compared to when it is not. These empirical results, obtained in the same sample employed by KNS, are difficult to reconcile with the implications of their model that (i) “expected returns must line up with the first few (high eigenvalue) PCs” (page 1193) and (ii) “either the SDF can be approximated well by a low-dimensional factor model with the first few PCs as factors, or the SDF cannot be volatile and hence Sharpe ratios only very small” (page 1203). This suggests that the cross-factor arbitrage that they rely on to obtain their implications is confronted by relatively large barriers. Instead, the outcomes support the conclusions that (1) allowing for time

⁹ The numerator provided by standard mean variance mathematics is a set of non-normalized weights. These are typically normalized by dividing by the sum of the non-normalized weights. However, since each factor and factor PC is a zero-net-investment long-short portfolio, we instead divide by the sum of the absolute values of non-normalized weights. This normalization ensures that SRs are positive (eliminating the need to work with squared SRs), but otherwise does not alter SRs as compared to the standard normalization or not normalizing at all (i.e., only using the numerator).

variation in factor premia enhances the ability to forecast the cross section of stock returns, and (2) more factor PCs are relevant when time variation is allowed for than when it is not.¹⁰

b. Broader Evidence: 212 Factors from CZ

KNS studied fifteen “anomaly” factors and twenty-five size and market-to-book factors for which return data was available during all months of their fifty-year sample period. Of course, the literature has collectively studied a much broader set of factors, some of which have available data for longer time intervals than others. Our rolling estimation approach requires only that factor return data is available during relatively short in- and out-of-sample estimation windows. Thus, though we study outcomes spanning calendar years 1926 to 2024, we can include the many factors for which return data is available over shorter time periods. The dates displayed on our Figures pertain to the end of the rolling in-sample estimation periods, but each analysis incorporates the subsequent thirty-six month OOS period.

We implement our rolling estimation approach using 212 factors derived from firm characteristics previously studied in the literature, including the 165 “clear predictors” and 47 “likely predictors” identified by Chen and Zimmerman (2022). We estimate OOS Sharpe ratios for portfolios formed from varying numbers of factor PCs to assess the extent to which the relevant economic information is or is not redundant across many factors.

We report in Panel A of Figure 2 annualized OOS Sharpe ratios for portfolios formed from the PCs of the CZ factors, when the end of the sixty-month in-sample estimation period ranges from June 1931 to December 2021. The yellow line displays Sharpe ratios for portfolios formed from the first five PCs (in line with the results of Hou, Xue, and Zhang, 2020 and KNS) and the blue line shows the average Sharpe ratio obtained across portfolios consisting of 6 or more PCs (from six to a maximum of fifty-nine). The orange line shows the Sharpe ratio obtained with the maximum number of PCs that can be

¹⁰ The conclusion that allowing for time variation in factor premia can improve outcomes is also supported by the results reported by Neuhierl, Randl, Reschenhofer and Zechner (2023). They show that factor return premia, including most notably those associated with value and firm profitability, can be forecast to a significant extent using variables such as prior factor returns and volatility.

constructed when using a sixty-month estimation window.¹¹ Table 1 reports time series averages of each of these Sharpe ratios, as well as differences in average Sharpe ratios and associated t-statistics.¹²

The data displayed on Panel A of Figure 2 and the summary statistics in Table 1 indicate that the CZ factors collectively have economically important out-of-sample predictive power for the cross-section of stock returns. The annualized OOS Sharpe ratios for portfolios formed from the first five factor PCs are mostly positive, average 1.28 for the full sample, and exceed two during some portions of the 1980s and 1990s. The blue line displays average annualized Sharpe ratios across six to fifty-nine PCs. These Sharpe ratios are virtually always positive, average 2.75 across months, and in the substantial majority of months are larger than those based on five PCs. The Sharpe ratio based on fifty-nine factor PCs averages 2.84 across months. These results reaffirm that observable factors have significant forecast power for the cross-section of returns.

The Sharpe ratios displayed on Figure 2 are generally larger than those displayed on Figure 1, implying that the larger set of CZ factors contain more information than the factors studied by KNS. Substantial time variation is readily observable on Panel A of Figure 2. For example, average OOS Sharpe ratios when employing the maximum number of PCs are most often in the vicinity of two to three during the 1960s, but often range from five to seven during portions of the 1980s and early 1990s. In Section IV we assess whether the degree of time variation is consistent with random variation around constant factor premia.

Results reported in columns (1) to (4) of Table 1 indicate that the difference between the time series average OOS Sharpe ratio obtained from the average across six to fifty-nine PCs and the time series average based on five PCs is 1.47, while the difference in the time series average Sharpe ratio based on

¹¹ The maximum is most often 59, but in some applications (e.g. when studying only “significant” factors, as described below) is less. The statistics literature has noted that PC estimates become noisy and inconsistent when the number of PCs estimated approaches the sample size. Such statistical noise and inconsistency would be expected to only degrade OOS performance. Our results nevertheless show that the higher-order PC estimates contribute to high OOS Sharpe ratios and therefore contain substantial and economically relevant information.

¹² Since outcomes are based on rolling sixty-month in-sample estimation, we employ Hansen-Hodrick standard errors with a bandwidth of sixty.

the maximum number of PCs and that based on five PCs is 1.56. Each of these differences is highly statistically significant, with t-statistics of 7.31 and 3.28, respectively. These outcomes imply that factor PCs beyond the first few contribute substantially to the ability to forecast the cross-section of stock returns, and therefore contain non-redundant information.¹³

KNS note that in-sample Sharpe ratios increase mechanically when portfolios are formed from a larger number of factor PCs. To verify that OOS SR do not increase mechanically, we also report in Table 1 the time series average of the maximum Sharpe ratio obtained across any number of PCs. This maximum Sharpe ratio exceeds the Sharpe ratio obtained with the maximum number of PCs in every rolling sixty-month interval, and averages 4.18 across months. The average number of PCs for the portfolio that gives the maximum out-of-sample Sharpe ratio is forty.¹⁴

While these outcomes can only be observed ex post, they confirm that OOS Sharpe ratios do not mechanically increase when more factor PCs are employed. Instead, the large maximum Sharpe ratios and the large numbers of PCs used to construct these portfolios imply that many factors contain economically relevant and non-redundant predictive information regarding the cross-section of stock returns.

Our findings contrast with the conclusions of recent papers that conclude that only relatively few factor PCs are relevant. Our conclusion differs mainly because our rolling estimation approach allows for flexible time variation in factor return premia in a way the prior studies do not.

¹³ We also assess the explanatory power of individual factor PCs, from the first to the fifty-ninth, by assessing Sharpe ratios for that PC factor alone. In Appendix Figure A1, we report the average (across months) in-sample and OOS Sharpe ratios for each factor PC. Consistent with the reasoning expressed by KNS (2018), in-sample Sharpe ratios increase almost monotonically for the higher numbered PCs. In contrast, OOS Sharpe ratios are nearly constant, equal to approximately 0.20 to 0.25 across all PCs from the first to the fifty-ninth. That is, the OOS explanatory power of higher numbered PCs is neither systematically greater nor smaller than that of the first few PCs. Table IA-4 of the internet appendix displays additional information regarding each both in-sample and OOS SRs for each individual factor PC from the first to the fifty-ninth.

¹⁴ Figure IA-2 in the internet appendix displays outcomes on this measure for each rolling window, and Table IA-3 displays related descriptive information.

c. Are outcomes stronger when considering only those factors that are significant in sample?

For the results reported to this point, we employ all factors with available data during the relevant estimation window. However, if factor premia vary across time but do not immediately revert to their long run means, then considering only those factors with significant recent explanatory power may reduce noise and improve the ability to forecast the cross-section of returns. However, improved OOS Sharpe ratios are not guaranteed, as forming PCs from fewer factors can reduce diversification benefits. To assess this issue, we classify a factor as having significant in-sample explanatory power in those cases where the t-statistic on the alpha estimated in an in-sample regression of factor returns on excess market returns exceeds two. A significant alpha in this regression implies that a factor is economically relevant in the sense that it has explanatory power for the cross-section of non-market returns.

Panel B of Figure 2 displays OOS Sharpe ratios that correspond to those in Panel A, but are obtained when PCs are formed from only those factors with significant in-sample alphas. The right panel of Table 1 (columns 6 to 10) reports the means and standard deviations across months of these Sharpe ratios. Column (3) of Table 1 present results when the number of PCs is selected (up to a maximum of 59) to match the number of significant factors with significant in-sample explanatory power, which we henceforth refer to as the number of “predicted PCs.”

The outcomes indicate that OOS Sharpe ratios for portfolios formed from the PCs of those factors with statistically significant in-sample alphas exceed those for portfolios formed from the PCs of all factors. In particular, the average annualized out-of-sample Sharpe ratio (across months) for portfolios formed from five PCs is 2.08 when the PCs are drawn from only significant factors versus 1.28 when the PCs are drawn from all factors. Corresponding outcomes for Sharpe ratios averaged across portfolios consisting of between six and the maximum number of PCs are 3.33 when PCs are constructed from only significant factors versus 2.75 when PCs are constructed from all available factors. In Section V we delve

more deeply into the empirical determinants of times series variation in the numbers of economically relevant factors.

Note that the SR obtained for the portfolio formed with the predicted number of PCs is mechanically equal to the SR obtained for the maximum number of PCs of the significant factors. However, the corresponding SR is meaningful when all factors are employed to create PCs. Strikingly, the mean OOS SR for the portfolio formed from the predicted number of PCs (column 3) is 3.30, which is greater than the corresponding average Sharpe ratios in columns (1), (2), or (4), (though it is less than the ex post maximum SR, in column 5). This indicates that the changes in the number of factors with in-sample explanatory power is informative as to the number of factor PCs with meaningful OOS explanatory power.

d. Outcomes with the simplest portfolio weights

KNS (2018) express the concern that constructing mean-variance optimal portfolios from large numbers of factor PCs could result in portfolios with extreme weights. Accordingly, they assert that “ignoring the small-eigenvalue PC portfolios” is appropriate for OOS evaluation. We next assess outcomes when OOS portfolios are formed from factor PCs by equal weighting. These results are informative as to whether our central results are robust to a simple method that rules out extreme weights by construction.

Figure 3 displays monthly outcomes that parallel those in Panel A of Figure 2, except that portfolios are equal-weighted across the indicated number of factor PCs. Table 2 presents corresponding time series means of these OOS Sharpe ratios. These outcomes also support the conclusions that factors contain significant forecast power for the cross section of stock returns, as OOS Sharpe ratios are economically large, and are greater when portfolios are formed from more than a few factor principal components. In particular, the average OOS Sharpe ratio for equal-weighted portfolios formed from five factor PCs is 1.43, while that for equal-weighted portfolios formed from the maximum number of factor PCs is 3.24.

The average OOS Sharpe ratios for equal-weighted portfolios as reported on Table 2 are similar in magnitude to those for mean-variance optimized portfolios as reported on Table 1. For example, considering the outcome averaged across six to the maximum number of PCs, the average Sharpe ratio for optimized portfolios on Table 1 is 2.75, while that for equal-weighted portfolios on Table 2 is 2.63. Thus, our central conclusions obtain even when portfolios are formed from factor PCs in the simplest possible manner.

e. Can OOS Sharpe ratios or the number of factor PCs that maximize OOS Sharpe ratios be forecast?

The number of factor PCs that lead to the highest OOS Sharpe ratio for any given thirty-six-month evaluation period can only be observed ex post. We assess the extent to which the number of factors with significant in-sample explanatory power is useful in terms of predicting OOS Sharpe ratios or the number of factor PCs that will be associated with the highest OOS Sharpe ratio.

We first provide on Panel A of Figure 4 descriptive information regarding the number of CZ factors with available data at various points in time, as well as the number of factors with statistically significant CAPM alphas within each rolling sixty-month-in-sample period. The number of significant factors in the pre-1960 data is always less than ten. However, more than one hundred factors had significant explanatory power during portions of the 1980s and 1990s. We also assess, for each rolling sixty-month in-sample period, the number of factor PCs required to explain 95% of the time series variation across the available CZ factors. To explain 95% of the in-sample variation in factor returns requires between twenty and thirty-five PCs for every rolling sixty-month window from the late 1950s through the end of the sample period. On Panel B of Figure 4 we display the number of PCs required to explain 50%, 60%, 75%, 90% of the variation in the available factors. Consistent with the results reported by Hou, Xue, and Zhang (2020), approximately fifty to sixty percent of the variation in the factors can be explained by a small number of PCs, ranging at various times from one to six. However, explaining a larger portion of the variation in the factors requires many more PCs.

The results reported in Tables 1 and 2 show that OOS Sharpe ratios are generally improved when PCs are formed from only those factors that are statistically significant. We now assess whether the number of significant factors contains predictive power regarding the portfolios that turn out to be optimal OOS. In particular, we estimate regressions where the dependent variables are, in turn, the OOS Sharpe ratio for the portfolio with the ex post maximum Sharpe ratio and the number of PCs used to construct that same portfolio. We assess results both when out-of-sample portfolios are formed from the PCs of all factors available in-sample (columns 1 and 2) and from the PCs of only the factors that are significant in-sample (columns 3 and 4).

The results, reported in Table 3 show that the number of factors that are significant in-sample does contain forecast power regarding optimal OOS portfolios. In particular, the number of significant factors positively predicts the number of PCs that comprise the portfolio with the highest OOS SR, when the PCs are constructed from all factors ($R\text{-squared} = 0.19$), and more notably when PCs are constructed from significant factors ($R\text{-squared} = 0.74$). In the latter case, the number of significant factors also predicts the maximum SR itself, with an $R\text{-squared}$ of 0.23. That is, during those periods when a larger number of factors can explain the cross-section of non-market returns in-sample period, OOS Sharpe ratios are maximized by employing a larger number of factor PCs, and the maximized OOS SR is larger.

The CZ factors were initially documented by widely disparate authors, which allows the possibility that some of the factors might be largely duplicative of the information contained in others. However, the results described here show not only that the factors collectively contain substantial information regarding the out-of-sample cross-section of returns, but that larger numbers of in-sample significant factors contain more information relevant to predicting the OOS cross-section of stock returns.

IV. Could the Observed Variation in Factor Premia Simply Be Random?

The literature has debated the extent to which the body of evidence that is often referred to as the “factor zoo” is reliable.¹⁵ We contribute to this discussion by assessing the extent to which factors have statistically significant explanatory power in subperiods before and after those examined in the studies that originally identified the factors. In doing so, we extend the related results reported by McLean and Pontiff (2016), Linnainmaa and Roberts (2018) and Ilmanen, Israel, Moskowitz, Thapar, and Lee (2021), who mainly studied factors on a univariate basis, by studying a larger set of factors on a multivariate basis.¹⁶ Also, while related studies have assessed whether factors do or do not have explanatory power for the pre- and post-sample periods as a whole, we assess the extent to which factor premia vary over time. We also subsequently present the results of numerous robustness checks, including outcomes obtained when we construct new factors from the WRDS financial ratios dataset, which is arguably immune to the data snooping criticism.

Figure 5 displays richly detailed information regarding the significance of each of the 212 CZ factors during periods before, during, and after the sample periods studied by the original authors. The figure includes one row for each factor, and a column for each sample month. Factors are sorted based on their unconditional CAPM alpha t-statistics during the original sample period, from lowest to highest. A given row and column contains a blue dot if the t-statistic on the factor’s alpha estimated in a CAPM regression over the prior sixty months is greater than two. In addition, each row contains a green dot that denotes the earliest data used in the original study that identified the factor, a red dot that denotes the latest data used in the original study, and a magenta dot that indicates the earliest date for which data is now available.

¹⁵ See, for example, Harvey, Liu, and Zhu (2016), Linnainmaa and Roberts (2018), Chordia, Goyal, and Saretto (2020), Hou, Xue, and Zhang (2020), Chen (2021), Chen and Zimmerman (2022), and Jensen, Kelly, and Pedersen (2021).

¹⁶ However, replication rates are not directly comparable across our study and theirs, as we focus on a set of 212 factors that were previously verified by Chen and Zimmerman to have significant explanatory power within the authors’ original sample periods. In contrast, only 85 of the 97 factors studied by McLean and Pontiff (2016) have an in-sample t-statistic greater than 1.50, and only 32 of the 36 factors studied by Linnainmaa and Roberts (2018) have an in-sample t-statistic greater than 1.96.

Two key points can be extracted from the information displayed on Figure 5. First, factors often display statistically significant explanatory power in data drawn from months both before and after the data used in the original study that identified the factor. In Panel A of Table 4 we report data on the frequency of such occurrences. If significance is assessed by a t-statistic on the alpha estimate of 1.96 or greater, over three quarters (79%) of the factors have significant explanatory power during at least one 60-month interval prior to the range of dates used in the original studies, and 92% have significant explanatory power during at least one sixty-month interval after the range of dates used in the original studies. If statistical significance is defined based on a larger t-statistic, the proportion of factors that are significant outside of the original sample period declines but remains large. For example, applying the t-statistic of 3.00, 56% of factors are significant during at least one earlier sixty-month interval and 67% are significant during at least one subsequent sixty-month interval. This out-of-sample evidence supports the reasoning that the factors' success in the original studies cannot be fully attributed to data mining or specification searches.

The second observation that can be gleaned from Figure 5 is that the statistical significance of individual factor alphas varies over time; in many cases a given factor is significant for periods spanning multiple years, loses significance for a time, and then regains significance. In Panel B of Table 4 we report evidence on the distribution of the number of non-overlapping periods, or "spells" of significance, for various t-statistic cutoffs. For example, relying on a t-statistic cutoff of 3.00, the cross-factor median number of significance spells is 3 per factor, while the cross-factor mean is 4.2 spells per factor. Panel C of Table 4 reports on the distribution of the duration of such spells. Once again based on a t-statistic cutoff of 3.00, the cross-factor median length of a significance spell is 15 months, while the cross-factor mean length is 23 months.

Of course, the statistical significance of individual factors could vary over time simply due to random noise in a stable economic environment. That is, a factor with a constant premium equal to zero or an economically modest quantity could be associated with significant estimates during some intervals and insignificant estimates during other intervals. Alternatively, the pattern could reflect that the number

of factors that earn a return premium, or the magnitude of such return premia, vary over time. We distinguish between these explanations in two ways. First, we use simulation methods to assess the distribution of the statistics reported on Panels B and C of Table 4 under the null hypothesis that factor premia are constant over time. Second, we present evidence in Section VII that assesses the extent to which variation in the number of significant factors is related to measures of changes in the economic environment.

To assess the distribution of the statistics reported on Table 4 under the null hypothesis that factor premia are constant over time, we rely on simulations, as follows. First, we estimate each factor's constant alpha, beta and residual volatility from a regression of its returns on the market excess returns. We then create a simulated time series of market returns calibrated to the sample mean and standard deviation of the market over our sample period and generate a simulated time series of returns for each of the 212 factors using a factor model that relies on the simulated market returns in combination with the estimated alpha, beta, and residual volatility for each factor. The length of each factor's simulated time series is matched to the number of sample observations for the factor return. Having done so we estimate rolling 60-month regressions of simulated factor returns on simulated market returns and obtain both the count and average length of significance spells for each simulated factor, when significance is assessed based on t-statistics ranging from 1.96 to 4.00. We compile the cross-factor average of the spell counts and spell lengths, corresponding to the sample data reported in Table 4. We repeat the simulation 2,000 times to obtain a distribution of the cross-factor average factor spell lengths and counts under the null hypothesis of constant factor premia.

Panel A of Figure 6 displays the simulated distributions for the cross-factor average of the average spell lengths, while Panel B displays corresponding cross-factor average spell counts. The red dashed lines denote outcomes from the actual sample data. This data shows that the statistics pertaining to the actual sample and reported on Table 4 are unlikely to be observed under the null hypothesis of constant factor premia. For each of the t-statistic cutoffs considered, the actual average spell length lies far in the right tail of, or entirely outside, the simulated distribution of spell lengths. That is, actual spell

lengths are longer than would be observed under the null hypothesis, as would be anticipated if premia varied through time without immediate reversion to their long run means.

The information on Panel B of Figure 6 displays a more complex pattern. The use of a higher t-statistic cutoff mechanically leads to fewer periods of significance, both in the sample data and in the simulated distribution obtained under the null. Note, though, that with high t-statistic cutoffs of 3.5 or 4.0, the actual average count of significance spells lies far in the right tail of the simulated distribution, while with low t-statistic cutoffs of 1.96 or 2.00, the actual average count of significance spells lies in the left tail of the simulated distribution. That is, simulated outcomes that were obtained under the null hypothesis are more sensitive to the t-statistic cutoff as compared to actual sample outcomes. Empirical estimates of economically modest and time-invariant return premia are likely to be recategorized as insignificant rather than significant as the t-statistic hurdle increases. In contrast, the t-statistic employed is of less relevance for factor premia that are economically large during some periods (as they are then more likely to remain significant even as the t-statistic threshold is increased) and close to zero during other periods (when they are likely to be insignificant even with low t-statistic thresholds).

On balance, the simulation outcomes displayed on Figure 6 imply that it is exceptionally unlikely that the sample data reported on Table 4 could simply reflect random variation under the null hypothesis of constant factor premia. Rather, the sample outcomes are consistent with those that would be anticipated if return premia associated with individual factors were economically large across some periods and near zero at other times. This finding contrasts with a perspective that has appeared in the prior literature, by which a return premium deemed to be anomalous should be permanently eliminated by arbitrage activity once the existence of the premium becomes broadly known. Rather, we document that factors often display post-sample spells of significance that occur even after periods of insignificance.

V. Are the Results Regarding the Importance of Allowing for Time Variation Robust?

The results reported in Tables 1 and 2 support the conclusion that allowing for time variation in factor premia substantially improves the OOS Sharpe ratios for portfolios formed from factor principal components. We next assess whether this conclusion is robust to several aspects of research design.

a. Alternative methods to extract factor principal components.

For all outcomes reported in Section III, we follow KNS (2018) in employing standard methods to extract factor principal components. In this section we assess if our central finding, that allowing for time variation in factor premia leads to the implication that OOS Sharpe ratios continue to improve as more than a few factor PCs are employed also applies when PCs are extracted while employing the methods of Lettau and Pelger (2020) or Bryzgalova et al. (2023). While standard PCA analysis focuses entirely on explaining the variance of returns, the Lettau and Pelger (2020) method also puts weight on the vector of mean returns, while the Bryzgalova et al. (2023) approach allows the targeting of specific cross-sectional and time-series objectives.¹⁷

Table 5 reports results that parallel those contained in Table 1. In particular, we report time series averages of OOS Sharpe ratios for portfolios formed from varying numbers of PCs that are extracted from the CZ factors using the methods of Lettau and Pelger (Panel A) and Bryzgalova et al. (Panel B). Results are reported both when all 212 CZ factors are employed, and when only employing the factors with significant explanatory power during preceding sixty months. As in Table 1, we report the outcome when

¹⁷ The key parameter for each method is gamma. We set gamma parameters to those used in Bryzgalova et al. (2023). In particular, to implement the Lettau and Pelger (2020) method, the time series gamma is 15 and the cross-sectional gamma is 0. We find similar results when instead setting the time series gamma to 10, which is the baseline employed in Lettau and Pelger (2020). To implement Bryzgalova et al. (2023) method, the time series gamma is 15 and cross-sectional gamma is 1. To isolate the effect of the PC extraction method, we continue to employ rolling 60-month in-sample estimation. In Appendix Table A2 we report results obtained when we follow Lettau and Pelger (2020) and Bryzgalova et al. (2023) in using rolling 240-month in-sample estimation. Here too, every t-statistic indicates that OOS Sharpe ratios are greater when more than five factor PCs are employed. It can also be observed that the OOS Sharpe ratios obtained by these methods are generally larger when the longer 240 month in-sample estimation period is employed. For example, the average SR obtained by the Bryzgalova et al. (2023) method using all factors and the maximum number of PCs is 2.95 with 240 month in-sample estimation, compared to 1.71 with 60-month in-sample estimation. This parallels the outcome obtained for our own method when we lengthen the in-sample estimation window, as demonstrated in Appendix Table A4.

employing the first five PCs, the average outcome obtained across six to the maximum number of PCs, the outcome when using the predicted number of PCs (the number of in-sample significant factors), and the outcome when employing the maximum number of PCs, and tests of the hypothesis that each mean differs from the outcome with five PCs.

The outcomes contained in Table 5 demonstrate that increasing the number of PCs beyond five significantly improves OOS Sharpe ratios in all cases. For example, employing the Lettau and Pelger method and all factors, the average OOS SR with the maximum number of PCs is 1.47, compared to 0.90 with five PCs. The t-statistic for the 0.57 average SR increase is 3.21. When the Bryzgalova et al. method is employed instead, the corresponding average OOS SR with the maximum number of PCs is 1.71, compared to 1.01 with five PCs. The t-statistic for the 0.70 increase in average Sharpe ratios is 5.24. It can also be observed on Table 5 that all OOS Sharpe ratios are larger when PCs are extracted from only those factors that are significant in sample, as compared to corresponding estimates obtained when using all factors. That is, our key findings that OOS performance is improved by allowing for more than five factor PCs and by focusing on the time-varying number of in-sample significant factors are robust to these alternative methods of identifying factor PCs, thereby underscoring the importance of accounting for time variation in factor premia via many PCs.

b. An Alternative Set of Factors

As noted, we study the 212 factors compiled by CZ. A natural question is whether studying alternative factors would lead to the same central conclusions. To assess this issue, we construct factors based on the sixty-eight ratios contained in the WRDS Fundamental Ratios dataset. Each CZ factor pertains to an underlying characteristic featured in a previous-published study. The WRDS dataset to our knowledge is immune to the potential concern that the financial ratios were defined as the result of data mining exercises related to return predictability. The WRDS dataset is therefore unlikely to have been affected by data snooping or specification searches that may arguably have been relevant for some of the papers underling the factors compiled by CZ.

To construct the sixty-eight series of factor returns from this dataset of financial ratios, we employ methods that parallel KNS (2020). In particular, we sort stocks based on each financial ratio and create an ordered weighting (a normalized ranking between -1 and 1) for each stock. Stocks with a weighting less than zero on a given ratio comprise the short leg for the new factor, while those with a weighting greater than zero comprise the long leg. Each factor return is obtained as the product of stock returns and weights.

Table 6 reports outcomes obtained when we implement in the alternative WRDS-based factors methods that correspond to those we employed for the CZ factors, as reported in Table 1. Here too, it can be observed that OOS Sharpe ratios increase when more than a few factor PCs are employed. In particular, the average OOS SR when the maximum number of PCs are extracted from the full set of WRDS-based factors is 1.85, compared to 1.13 when five PCs are employed. The t-statistic for the increase in the average Sharpe ratio is 3.96. Results obtained when we extract PCs from those factors with significant explanatory power over the preceding sixty months are somewhat weaker, reflecting in part that the smaller sample of 68 WRDS-based factors provides fewer significant factors as compared to the 212 CZ factors. However, results are still improved, as the average SR is 1.63 when the maximum number of PCs (which is often less than fifty nine) are employed, as compared to 1.37 when five factors are employed. The facts that (i) we estimate economically substantive OOS Sharpe ratios, and (ii) OOS Sharpe ratios are larger when more than five factor PCs are employed even in a set of factors constructed independently of those in the CZ database indicates that the key findings of this paper are not merely artifacts of data snooping or selective publication.

c. Are the Results Driven by Small or Low Volume Stocks?

The weighting on individual stocks within the CZ factors is determined by the authors of the original papers that documented each factor's effectiveness. The majority of these authors elected to weight stocks equally, which raises the possibility that the factors' effectiveness may be disproportionately attributable to small capitalization or illiquid stocks.

We first assess the effect of reconstructing the CZ factors while excluding micro capitalization stocks. In particular, in each time period, we exclude any stock with a beginning-of-month market capitalization that is less than the 20th percentile of the market capitalization distribution for New York Stock Exchange listed stocks. Since Nasdaq stocks tend to be smaller than NYSE stocks, this eliminates the majority of sample stocks. The outcomes obtained for this modified set of CZ factors are reported in Panel A of Table 7. Here too, we observe that Sharpe ratios are economically large, and all average Sharpe ratios increase significantly when more than five PCs are employed. When using all available factors, the average OOS SR is 1.08 with five PCs, 2.05 with the predicted number of PCs, and 1.72 with the maximum number of PCs. The t-statistics for the 0.97 and 0.64 improvements in the average Sharpe ratios are 5.44 (column 3) and 3.38, (column 4), respectively. When we consider only those factors with significant explanatory power over the prior sixty months, the improvement in the average OOS SR when moving from five to the maximum number of PCs is 0.42, with a t-statistic of 2.89.

We then repeat this exercise, while excluding stocks with average daily turnover in the prior month that is less than the 20th percentile of the turnover distribution for New York Stock Exchange listed stocks. Outcomes are reported in Panel B of Table 7. When employing all factors, the OOS SR with five factor PCs averages 1.38, while the average Sharpe ratios with predicted number of factors and the maximum number of factors are 2.78 and 2.34, respectively. The t-statistic for the 1.39 (column 3) and 0.96 (column 4) improvements in the average OOS SR are 6.59 and 2.86, respectively. When employing only statistically significant factors the corresponding improvements are 0.33, with a t-statistic of 2.68. These results demonstrate that our central findings are not simply attributable to small capitalization or low volume stocks.

d. Are the Results Overturned when Allowing for Trading Costs?

We next assess whether our central results are overturned when we make reasonable allowances for trading costs. To do so, we follow the methods and employ the cost estimates of Kelly et al. (2023). In particular, we focus on the trading implied by month-to-month changes in the weights on individual

stocks within the portfolio that maximizes the in-sample SR (these weights also define the portfolio for which we assess OOS SRs). More specifically, following Kelly et al. (2023), we first assess each stock's "evolved" weight, as implied by the product of prior month weights and current month gross returns. The difference between each stock's actual and evolved weight reveals the trading required to achieve the current month weight. Having identified the implied amount of trading, and following Kelly et al. (2023), we deduct ten basis points times the absolute difference between the actual and evolved weight for non-microcap stocks, and twenty basis points times for microcap stocks. As a sensitivity test, we assess outcomes when these costs are doubled to twenty and forty basis points, respectively. In addition to these trading costs, we further deduct for the absolute value of any short position costs of the same magnitude (10 or 20 basis points in the base case, and 20 or 40 basis points as a sensitivity test).

For this analysis, we select the number of PCs during each OOS period to be equal to the number of significant factors during the prior thirty six months. We report results on a rolling one-month basis. That is, portfolio weights in each month $t+1$ are those that define the mean-variance optimal portfolio during the in-sample months $t-59$ to $t = 0$, implying that trading is required each month. Results are based on the 1984 to 2021 period, to correspond to the asset pricing comparisons reported in Section VI.

Table 8 reports average OOS Sharpe ratios with and without allowance for transaction costs. For comparison, we report the SR for the value-weighted portfolio of all CRSP common stocks during the sample period, which was 0.53. With no allowance for transaction costs, the mean OOS Sharpe ratio for our rolling one-month approach is 2.30. Allowing for transactions costs naturally decreases SRs. In the base case with 10 basis point trading and shorting costs for stocks that are not microcaps and 20 basis point trading and shorting costs for microcaps, the average OOS SR is reduced to 1.77. When transaction and shorting costs are doubled to 20 and 40 basis points, the average SR is reduced further to 1.01. These average Sharpe ratios continue to substantially exceed the value-weighted market SR during the sample period.

In our view, evidence of time variation in factor premia would be important even if reasonable estimates of transaction costs had fully eliminated the improvement in portfolio Sharpe Ratios, as it is of inherent interest to understand the determinants of the cross-section of expected stock returns and firms' costs of equity capital. The fact that allowing for reasonable estimates of trading costs still leads to OOS Sharpe ratios that substantially exceed the SR for the value-weighted market portfolio during the sample period only enhances the interest and importance of the results presented here. However, it should also be noted that our results do not necessarily imply “money left of the table,” as capturing the improved Sharpe ratios would have required real-time knowledge of the time variation that we now document in the historical data. That is, informational and learning costs comprise an additional barrier to effective arbitrage.

e. Additional Robustness Tests

In the Appendix to this paper we report on two additional robustness tests. First, we assess outcomes when we allow for time variation in factor premia that stems from changes in factor betas in particular. Specifically, we use the IPCA framework of Kelly et al. (2019) to construct seventy-five latent factors from the firm characteristics that underly the CZ factors.¹⁸ Having done so, we extract varying numbers of PCs of the latent factors, and form OOS portfolios in the same manner as previously described for the results reported in Table 1. Outcomes are reported in Appendix Table A3. Here too, we find that OOS Sharpe ratios increase significantly when the number of PCs is increased beyond five. That is, our results are robust to studying latent rather than observable factors, when allowing for time variation in factor betas in the manner specified by Kelly et al. (2019).

¹⁸ The Kelly et al. IPCA method requires the set of characteristics employed in the analysis to be fixed throughout the sample. Further, stocks can only be included in the analysis at a given date if they have non-missing observations for all employed characteristics. These requirements create a tradeoff, as employing more characteristics reduces both the length of the time series and the number of stocks that can be employed. Figure A2 in the appendix illustrates the tradeoff between the number of characteristics and the number of stock/month observations that can be employed in the CZ sample. Our selection of 75 characteristics allows for a sample size that is similar to that employed by KPS, while larger numbers of characteristics would have implied steep drop-offs in the available sample size.

Second, we assess the effect of altering the length of our in-sample and OOS estimation windows from the 60-month (in sample) and 36-month (OOS) windows employed as our baseline analysis. In particular, we consider alternate in-sample windows of 36 and 120 months, and alternate OOS windows of 60 and 120 months. The outcomes, reported in Appendix Table A4, show that our central results are robust, as all OOS Sharpe ratios are substantially larger with more than five PCs.

VI. Does Allowing for Time Variation Also Lead to More Accurate Pricing?

All of the analyses on which we have reported to this point focus on OOS Sharpe ratios for portfolios constructed from factor PCs. Sharpe ratios are of inherent interest to traders, and increases in Sharpe ratios imply greater ability to explain the cross-section of stock returns. Further, as Hansen and Jagannathan (1997) show, higher Sharpe ratios imply that portfolio returns are closer to the single SDF that fully explains the cross-section of returns.

We next turn to formal tests of whether allowing for time variation, and in particular the resulting insight that it is useful to employ a relatively large number of factor PCs, leads to more accurate asset pricing as compared to alternative specifications. We employ as test assets the 212 CZ factors themselves. More specifically, we compare outcomes from our rolling sixty-month estimation approach to those from canonical models, including the Fama and French (1993) three factor model, the Fama and French (2015) five factor models, the Hou, Xue, and Zhang (2015) Q-factor model.

Our intent is to evaluate the importance of allowing for time variation in factor return premia, not to claim that our relatively simple rolling-estimation approach necessarily leads to the best available asset pricing model. For that reason, we employ for the canonical models the split sample approach of KNS, which does not accommodate time variation in factor premia. In particular, we estimate in-sample mean variance portfolio weights using data from 1963 to 1983, while evaluating OOS Sharpe ratios outcomes using data from 1984 to 2024. We also compare outcomes from our rolling estimation method to those obtained from the five-PCA approach of KNS. We implement our own rolling 60-month in-sample

approach, with the number of PCs set to the time-varying number of significant in-sample factors, to estimate time-varying mean-variance optimizing portfolio weights, and evaluate outcomes of rolling one-month OOS periods, as in Table 8.

As a baseline, we report in Table 9 the OOS Sharpe ratio for each approach. These are 0.68 for the FF3 model, 1.58 for the FF5 model, 1.42 for the Q-factor model, 1.32 for the five PC approach, and average 2.30 for our rolling window method. All of these substantially exceed the 0.53 SR for the value-weighted portfolio of CRSP common stocks during the sample period, indicating that each method has explanatory power for the cross section of stock returns.

We also report in Table 9 three measures of how well the SDF produced by each model explains the cross-section of returns to the 212 CZ factors themselves. These include the Gibbons, Ross, and Shanken (GRS, 1989) test statistic, which evaluates the hypothesis that alphas obtained in regressions of factor returns on candidate SDFs are jointly equal to zero, the Hansen and Jagannathan (1997) distance, which assesses measures the magnitude of the difference between the candidate SDF and the SDF that, ex post, fully explains the cross-section of factor returns, and the cross-factor average absolute alpha from the regression of factor returns on candidate SDF returns.

The results of this exercise can be stated succinctly. The SDFs produced by the rolling window estimation procedure in combination with a relatively large number of factor PCs consistently provides the most accurate asset pricing results. Time varying estimation results in the lowest GRS test statistic, 4.98, compared to GRS statistics ranging from 8.72 to 10.31 for the models that do not allow for time variation. The HJ distance is 0.108 for our time-varying approach, as compared to outcomes ranging from 0.123 to 0.133 for the canonical models without time variation. Finally, the mean absolute alpha is 19.3 basis points for the time varying approach, as compared to outcomes ranging from 28.6 to 42.9 basis point per month for the models without time variation.

That is, the main insights of our paper hold not just for Sharpe ratios, but also for the pricing of the CZ factors themselves. In our view this provides compelling evidence of the importance of

accounting for time variation in factor premia and including more than the recently-recommended five PCs in the pricing portfolio.

VII. The Role of Economic Complexity

We next assess whether the time variation in the number of economically relevant factors is related to the complexity of the economic environment.¹⁹ We focus on outcomes for the 1968 to 2021 period, during which we can construct a larger set of such measures.

We consider the potential role of the number of firms traded in the public markets. We reason that substantial changes in the number of publicly traded firms attributable, for example, to surges in IPO activity, firm failures, or merger waves, are likely to be accompanied by shifts in the types of firms available for public investment. Consistent with this view, Fama and French (2004) show that the characteristics of firms newly listed on major U.S. stock markets varies over time. Canonical models would not predict that numbers of firms would be significant predictors. Rather, new stocks with unique characteristics would only require estimation of their potentially distinct factor betas, not the use of additional factors.

We consider the potential role of firm characteristics. Fama and French (2001) suggest that the disappearance of dividend-paying firms reflects the changing characteristics of publicly traded firms. To capture this aspect of variation in firm types, we compute the proportion of dividend-paying common stocks as the number of firms paying at least one cash dividend in the previous 12-months relative to the total number of common stocks. We also consider the possibility that the number of significant factors may be related to market liquidity and to general economic complexity. To the extent that factor premia arise because investors are unable to profitably trade to eliminate mispricing, we should observe that more

¹⁹ The variables we employ may tend to be persistent, and changes in unobserved variables that are correlated with those we are able to measure may also be persistent. For these reasons, our regression outcomes do not have causal interpretation, but rather indicate correlation between numbers of significant factors and these observable proxies. Future research may be able to identify superior measures of economic complexity and/or identify exogenous shifts to facilitate the exploration of causal effects.

factor premia are significant when markets are less liquid. To assess this possibility, we compute on a monthly basis the average across stocks of the Amihud (2002) illiquidity measure. As a proxy for general economic complexity, we use the Economic Complexity Index constructed by Simoes and Hidalgo (2011), which they describe as a measure of “the relative knowledge intensity of an economy.”

We consider also the potential role of diversity in firm characteristics. To do so, we first compute the cross-sectional standard deviation for each of the 212 characteristics within each month. We then rescale each of these measures such that the time series mean is zero and the time series standard deviation equals one. We compute the sum of these standardized volatility measures across characteristics within each month. The result can be interpreted as a measure of the cross-sectional dispersion in those characteristics observable for the available sample of firms in each month. If newly listed firms were like the typical existing firm in terms of observable characteristics the diversity measure would decline rather than increase as more firms listed. An increase in the cumulative dispersion in characteristics across firms, in contrast, indicates that the underlying firms themselves are becoming increasingly differentiated. To accommodate the “burn in” period between the addition of a firm and the time when characteristics become observable to investors, we focus on cross-sectional variation in characteristics in month $t+36$ to measure firm characteristic diversity as of month t .²⁰

Finally, we include as control variables measures of concurrent economic conditions. We conjecture that the business cycle may be relevant, both because of potential variation in the magnitude of return premia and due to changes in firm types, with economic expansions characterized by high rates of firm entry and recessions more likely to involve net exit by firms. To capture these effects, we rely on an indicator variable equal to one for recession months, as defined by the National Bureau of Economic

²⁰ Figure IA-1 in the internet appendix displays the average number of characteristics that can be computed, delineated by the number of months since the firm initially appears in the database. In the first few months, less than twenty characteristics can be computed, on average. Thirty-six months after listing, approximately one hundred characteristics can be computed. This rapid growth reflects the fact that many characteristics require prior accounting statement data (often sparsely collected at the beginning of a firm’s public life) and prior return history. However, the fact that a given characteristic cannot yet be computed by an econometrician need not imply that market participants are unaware of the characteristic.

Research, and the unemployment rate reported by the US Bureau of Labor Statistics. We also consider two interest rate series, the ninety-day Treasury-bill rate and the spread between the 10-year treasury note yield and the ninety-day rate. Interest rates potentially capture the effects of monetary policy and funding conditions. The unemployment rate, the fed funds rate and treasury yields are all obtained from the Federal Reserve Economic Data (FRED) website.

We report in column (1) of Table 10 outcomes obtained in a multivariate time series regression where the dependent variable is the number of significant factors during months $t - 59$ to t on time t outcomes for all of the variables described above. Contrary to our conjectures, coefficient estimates on the percentage of firms that pay dividends and the Amihud liquidity measure are not statistically significant. However, all of the other variables have significant explanatory power for number of significant factors. The regression adjusted R^2 statistic is 0.79, indicating that these variables collectively explain the majority of the variation in the number of significant factors.

Coefficient estimates for the macroeconomic control variables are statistically significant. Estimates on the recession indicator and the unemployment rate are both negative, consistent with the reasoning that recessions are likely to discourage entry, or encourage exit, by more complex firms, and also with the studies that documenting increases in return correlations during downturns (e.g., Ehling and Heyerdahl-Larsen, 2017). Coefficient estimates on estimates on both interest rate variables are positive and significant.

The coefficient estimate on the number of publicly listed firms is positive and significant in the multivariate regression. In column (2) of Table 10 we report the outcome when the number of publicly listed firms is the only regressor, to emphasize the univariate regression adjusted R^2 of 0.51, implying that the number of listed firms alone explains half the variation in the number of factors with significant explanatory power. The fact that more factors have significant explanatory power at times when more firms are listed is consistent with the reasoning that the firms that enter and depart the CRSP database differ from other firms in that they are exposed to differing sources of priced risk, rather than simply having differential exposures to a fixed set of priced systematic risks.

The coefficient estimates on the economic complexity index is positive and significant, consistent with the reasoning that the number of significant factors increases with complexity. The coefficient estimate for the diversity of firm characteristics is also positive and statistically significant.²¹ In column (3) we report results when economic complexity is the only explanatory variable and in column (4) we report outcomes when the diversity of firm characteristics is the only explanatory variable. Each of these also has strong explanatory power, with adjusted R^2 statistics of 0.21 and 0.29, respectively. That is, more factors are significant during those periods when there is greater economic complexity and more cross-sectional variability in firm characteristics. These are likely to be periods of greater uncertainty as to the determinants of expected returns. On balance, the results reported in Table 10 indicate that time variation in the number of significant factors is systematically linked to variation in macroeconomic conditions, economic complexity, the number of listed firms, and observable diversity in firms' characteristics.

VIII. Conclusions

The reasoning that only a few factors or factor PCs should be necessary to explain the cross section of mean returns is attractive in part because parsimony is desirable. So, should the fact that the literature shows that hundreds of observable factors have explanatory power for the cross-section of stock returns be viewed as a collective failure?

We think not. Financial markets and the broader economy are complex and dynamic. The characteristics of the firms that are available for investment can change over time as existing firms evolve and new firms are listed or delisted. Investors are diverse in terms of their investment horizon and objectives. Some investors trade on their own account, while others rely on professional managers whose

²¹ The positive relation between characteristic diversity and factor premia increase is not merely mechanical. Periods with larger cross-sectional dispersion in a given characteristic, e.g., firm size, need not mechanically display a larger long-short premium for the associated factor. Extant asset pricing models imply that such premia depend on the covariation of returns with economic aggregates, not with characteristic magnitudes. Further, even if returns are linked to characteristics in an anomalous manner, there is no a priori reason to believe such relations to be time-invariant.

strategies can be affected by agency issues related to their compensation. Return premia have been shown to depend on intermediaries' funding liquidity, leverage, and balance sheets, as well as on the state of the economy. The identity of the marginal investor can differ across stocks, and in a given stock, can vary through time. Arbitragers seek to exploit any inconsistencies in cross-sectional pricing, but face costs and other barriers that are only accentuated by the need to learn about time-varying parameters. In short, it is unclear that return premiums in actual capital markets are necessarily governed by only a small and time-invariant set of factors. Indeed, new factors may be necessary to price asset returns as the economy and its degree of complexity continue to evolve.

Cochrane (2011) observes that much of the variation in price-to-dividend ratios is attributable to changes in discount rates, i.e., expected returns. Prices are determined in market trading, based on the interaction between buy and sell orders. Cochrane (2022, page 31) observes that “the standard models ... do not produce a hundredth of the observed trading volume.” It follows, in our view, that the determinants of expected returns are not necessarily confined to those predicted by the same standard models, and can vary as market conditions, the economic environment, and other motivations for trading change. The need to be mindful of the possibility of collective data mining and joint hypothesis testing notwithstanding, these considerations support allowing the data to speak.

Fama (1998) articulated the advantages of studying observable rather than latent factors when seeking to understand the determinants of cross-sectional variation in expected returns, and we focus our analysis there, showing that observable factors identified in the literature have substantial forecast power for the cross-section of stock returns. Out-of-sample Sharpe ratios for portfolios formed from the PCs of in-sample factors are larger when time variation in factor premia is accommodated by means of rolling window estimation and when focusing on those factors with statistically significant in-sample explanatory power. Out-of-sample Sharpe ratios are also larger when more factor PCs are used to form portfolios, implying that the factors identified by prior researchers are to a substantial extent not redundant of each other. Further, the conclusions of existing studies that assert that only a few factor PCs are relevant are altered when the only change in research design is to allow for time variation in factor premia.

We use simulation methods to show that neither the average number of periods where a factor is significant nor the average duration of significance is consistent with the null hypothesis that factor premia are constant over time. We further show that the number of significant factors varies with measures of economic complexity and firm diversity. Our findings support the reasoning that newly listed firms systematically differ from existing firms in terms of systematic risks relevant to investors. Finally, the finding with respect to diversity of firm characteristics suggests that more factors are relevant when firms themselves are more distinct.

Our central finding, that allowing for time variation in factor premia leads to the implication that more than a few factor PCs are relevant in explaining the cross section of stock returns, is robust. The findings persist even when (1) we extract factor PCs by methods that consider the mean as well as the variance of factor returns, as in Lettau and Pelger (2020) or Bryzgalova et al. (2023), (2) we reconstruct factors to exclude small capitalization or low turnover stocks, (3) we rely on factors constructed from the WRDS financial ratios instead of factors discovered by prior researchers, and (4) we employ the IPCA method of Kelly et al. (2019) to construct latent factors that incorporate time variation in betas. We also show that allowing for the trading and shorting costs reduces, but does not eliminate the excess of the OOS Sharpe ratios we document compared to the overall market SR during our sample period. Further, we show that the SDF estimated by our time varying approach leads to improved asset pricing performance when compared to canonical asset pricing models without allowance for time variation, including the Fama and French (1993) three factor model, the Fama and French (2015) five factor model, the Hou, Xue, and Zhang (2015) Q factor model, as well as the five PC model of KNS (2018).

Our findings imply that a time-varying number of non-redundant factors are required to price the cross-section of returns as the economy evolves and diverse firms are listed and delisted. In such a dynamic economy, a factor can be significant in explaining returns during some periods but not others. This suggests that insignificant post-publication outcomes need not imply that a factor was unpriced in the original sample period nor that it will necessarily remain unpriced in the future.

The dangers of collective data snooping notwithstanding, our findings suggest that multiple and time-varying factors may be required to price the cross-section of returns as the economy continues to evolve dynamically and firms with differing characteristics are listed and delisted. Further, the fact that we document economically large out-of-sample Sharpe ratios that increase when portfolios are formed from large numbers of factor PCs supports the reasoning that arbitrage across factors is more difficult and costly than previously recognized, in part because of the need to identify in real time factor premia that are not constant.

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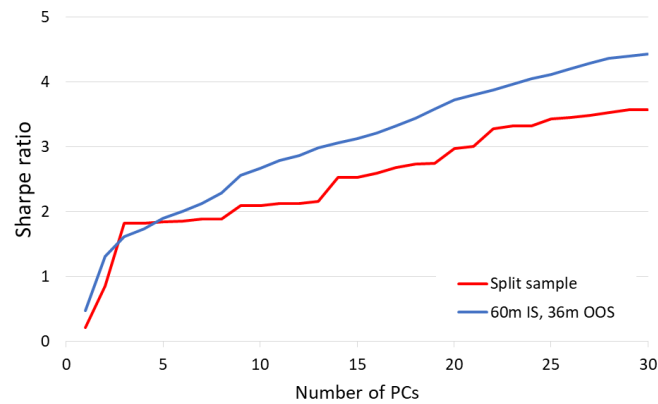
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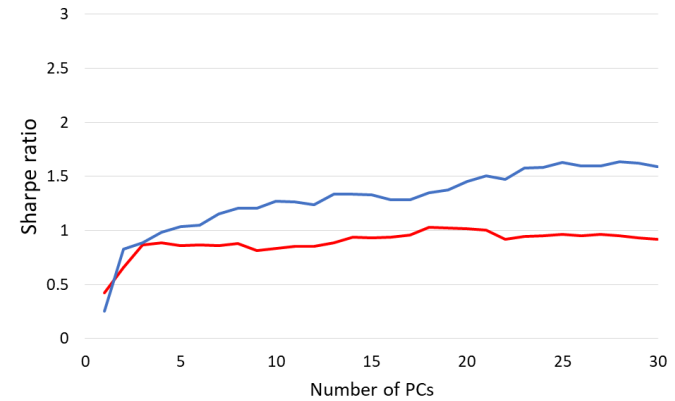
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Figure 1. Effect of PCs' estimation and forecasting horizons on out-of-sample Sharpe ratios. This figure shows the average annualized Sharpe ratios of portfolios formed from increasing numbers of PCs for different estimation windows of two sets of test assets from 1965-2015. Panel A shows the averages of rolling in-sample Sharpe ratios obtained from forming optimal portfolios consisting of different numbers of PCs extracted from the short and long legs of 15 anomalies using in Kozak, Nagel and Santosh (2018). Panel B shows the averages of rolling out-of-sample Sharpe ratios for the same assets. Panel C shows the averages of rolling in-sample Sharpe ratios for the 25 size and book-to-market portfolios. Panel D shows the averages of the corresponding out-of-sample Sharpe ratios for the same assets. The PCs are computed on a rolling monthly basis for different estimation windows. The out-of-sample portfolios are constructed using the in-sample optimal weights. The solid red line shows the Sharpe ratios where the sample is split in half and replicates Kozak, Nagel and Santosh (2018) using the first half as in-sample and the second half as out-of-sample. The blue line shows the Sharpe ratio where the in-sample window is 60 months and the out-of-sample window is the 36 months immediately following the in-sample window. For consistency with the split-sample analysis, the averages of the out-of-sample Sharpe ratios from the rolling sample are computed only using those out-of-sample windows that fall in the second half of the split sample.

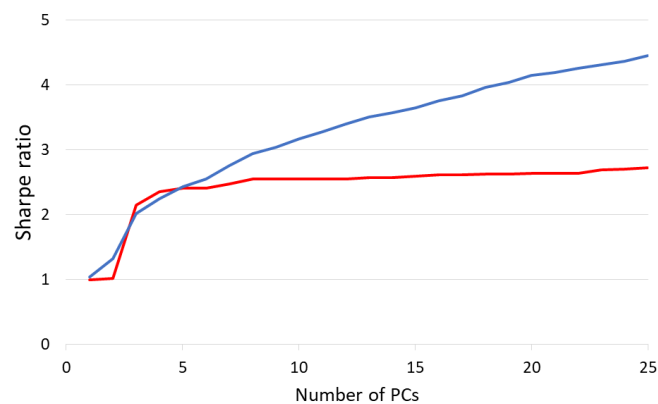
Panel A: In-sample Sharpe ratios – short/long legs of 15 anomalies



Panel B: Out-of-sample Sharpe ratios – short/long legs of 15 anomalies



Panel C: In-sample Sharpe ratios – 25 Size/BM



Panel D: Out-of-sample Sharpe ratios – 25 Size/BM

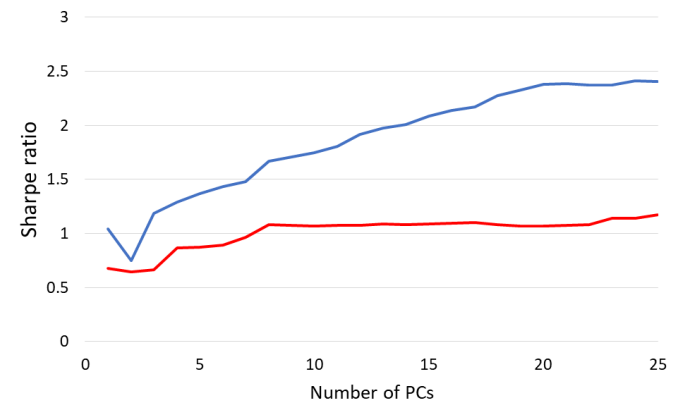
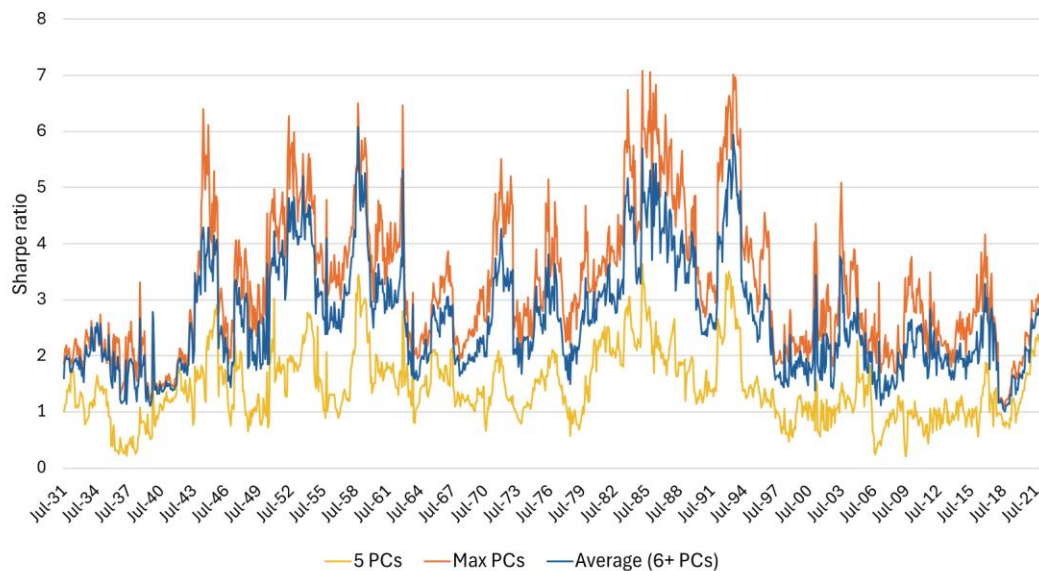


Figure 2. Out-of-sample Sharpe ratios of portfolios constructed from principal components. These figures show the annualized out-of-sample (subsequent 36 month) Sharpe ratios of portfolios formed from varying number of principal components. For each month t , we compute the factor loadings for the in-sample principal components for two sets of factor returns from $t - 59$ to t . We then use the factor loadings to form out-of-sample principal components from $t + 1$ to $t + 36$. We present 3 different Sharpe ratios formed from portfolios of PCs: 1) the SR from a portfolio of the first 5 PCs, 2) the average SR across portfolios formed by increasing numbers of PCs from six to fifty-nine PCs, and 3) the SR from a portfolio of the maximum number of principal components. Panel A shows results for PCs calculated from the set of all factors in our sample. Panel B shows results using PCs calculated for the subset of significant factors. See Section III for the full methodology. The missing points in Panel B are due to the lack of significant factors in the in-sample period.

Panel A: Out-of-sample Sharpe ratios of PC portfolios computed for all factors.



Panel B: Out-of-sample Sharpe ratios of PC portfolios computed from significant factors.

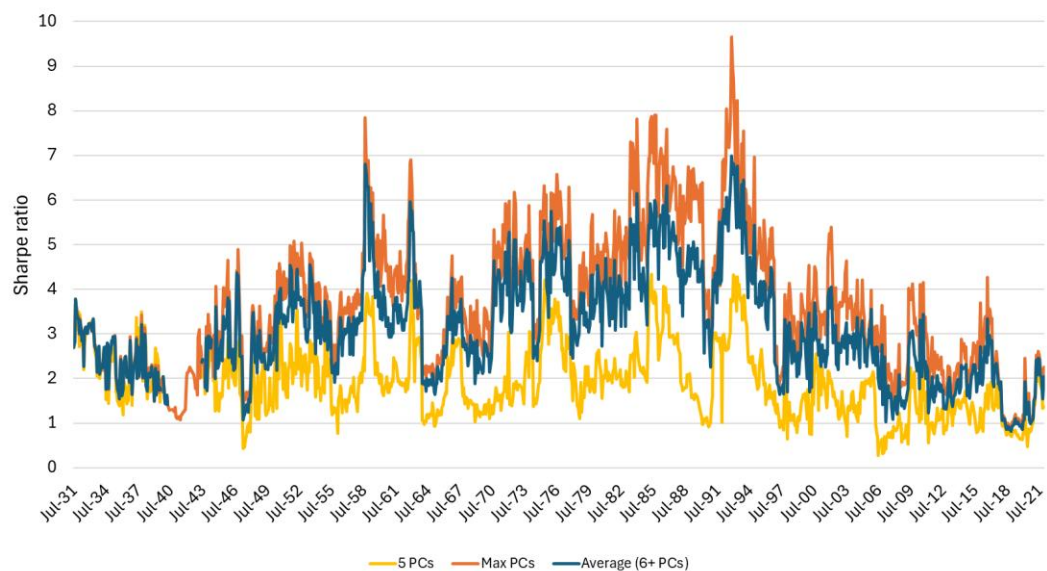


Figure 3. Out-of-sample Sharpe ratios of equal-weighted portfolios constructed from principal components. These figures show the annualized out-of-sample (subsequent 36 month) Sharpe ratios of portfolios formed from varying number of principal components for the set of all factors in our sample. For each month t , we compute the factor loadings for the in-sample principal components for two sets of factor returns from $t - 59$ to t . We then use the factor loadings to form out-of-sample principal components from $t + 1$ to $t + 36$. We present 3 different Sharpe ratios formed from portfolios of PCs: 1) the SR from a portfolio of the first 5 PCs, 2) the average SR across portfolios formed by increasing numbers of PCs from six to fifty-nine PCs, and 3) the SR from a portfolio of the maximum number of principal components. See Section III for the full methodology.

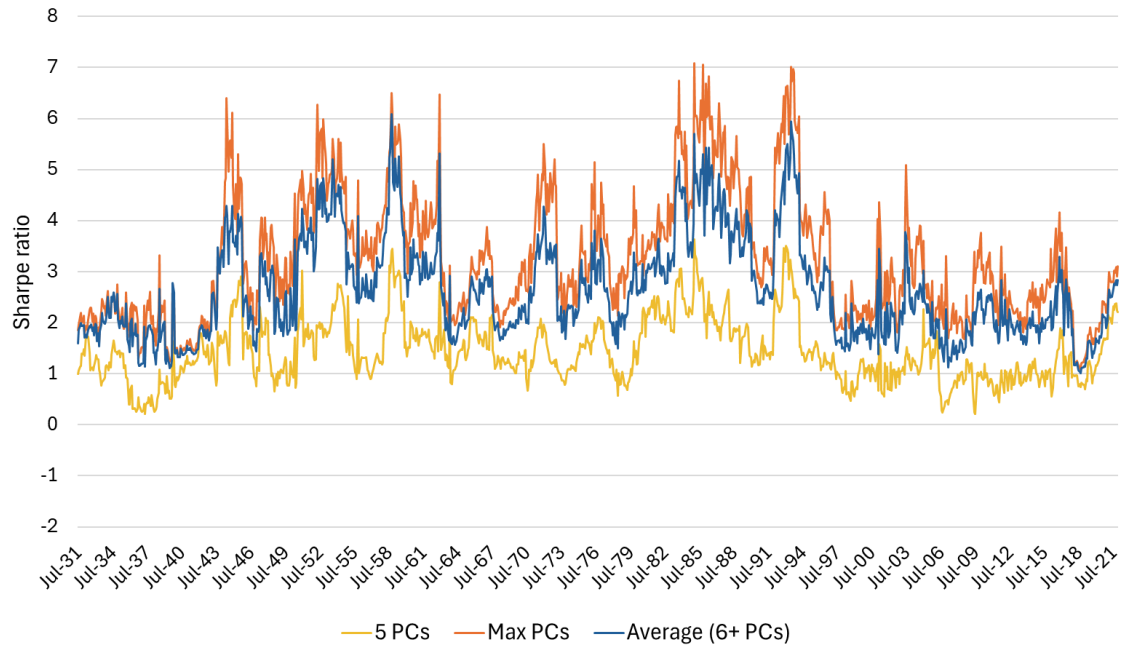
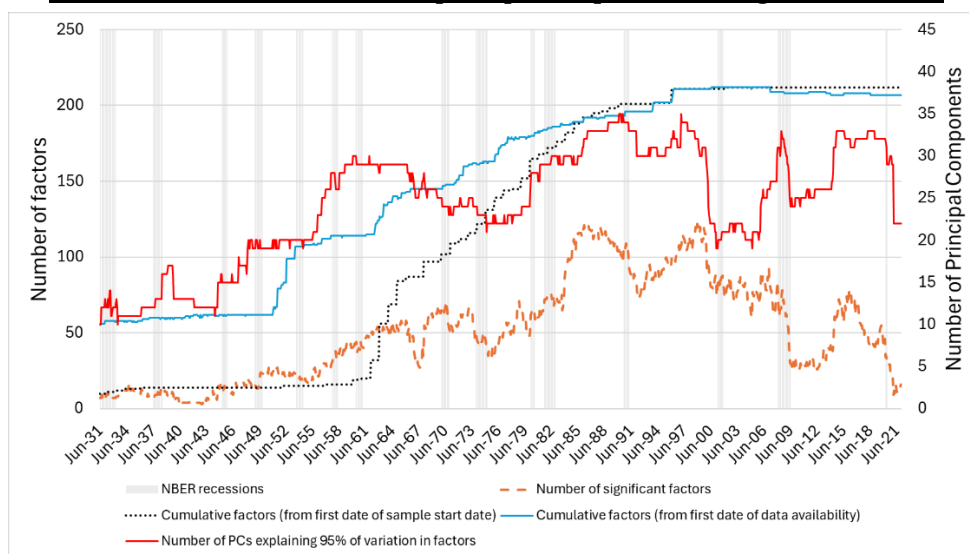


Figure 4. Time series variation in the number of principal components. This figure shows the number of principal components that explain the variation in a large number of factors documented in the finance literature. For each month t , we compute the in-sample principal components for a set of factor returns from $t - 59$ to t . We also compute the number of factors that have a significant (t -statistic > 2.00) CAPM alpha over the same in-sample period. The sample of factors comes from the set of 212 “clear” and “likely” predictors provided by Chen and Zimmermann (2022) from 1931 to 2024. The dotted black line in Panel A is the cumulative number of factors incremented at the date of each factor’s first available return based on the time period of the data used in the original paper’s sample. The solid blue line in Panel A is the cumulative number of factors incremented at the date of each factor’s first available return given the data available today. The solid red line of Panel A shows the number of principal components required to explain 95% of the cumulative variation in the factor returns from $t - 59$ to t . The dashed orange line shows the number of significant factors over time. Panel B shows the amount of variation in the factor returns explained by a given number of principal components. The grey vertical bars represent periods of NBER-defined recessions.

Panel A: Time series variation in principal components and significant factors



Panel B: Cumulative variation explained by principal components of all factor returns

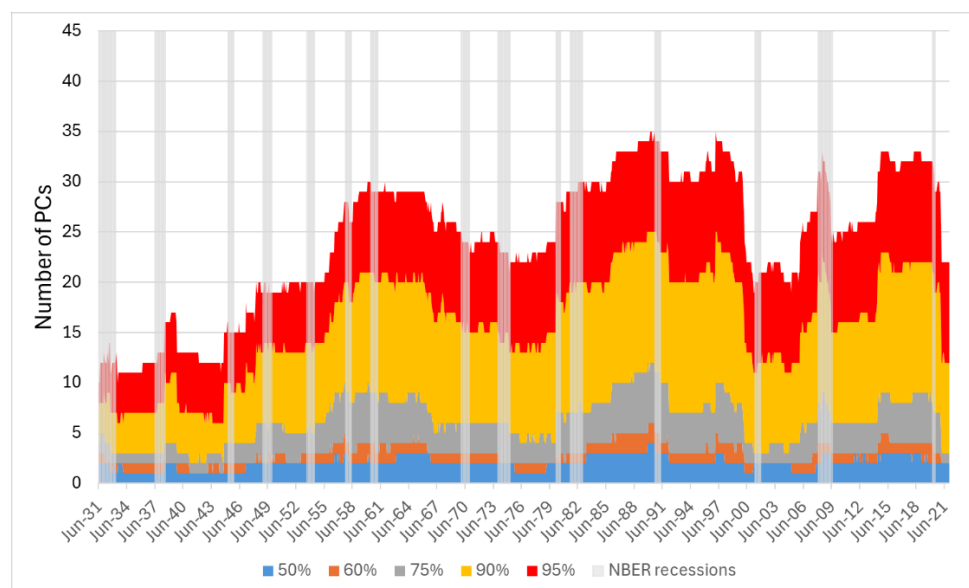


Figure 5. Time series variation in individual factors' significance. This figure shows whether a factor at a specific date has a statistically significant CAPM alpha over the preceding 60 months for the full sample of factors from 1931-2024. For each month t , we regress each factor's monthly returns from $t - 59$ to t on the market's monthly excess returns to obtain each factor's CAPM alpha. A factor is significant at month t if the t -statistic of its CAPM alpha exceeds 2.0. Factors must have 60 non-missing returns over the alpha estimation period. Each horizontal series represents a different factor with the blue dots signifying a month in which the factor is significant over the previous 60 months. A green dot denotes the earliest data used in the original study that identified the factor. A red dot denotes the latest data used in the original study. A magenta dot indicates the earliest date for which we are able to estimate the factor's alpha based on data now available. The factors are sorted in increasing order of the unconditional t -statistic of their unconditional CAPM alpha during the original sample period. The grey vertical bars represent periods of NBER-defined recessions.

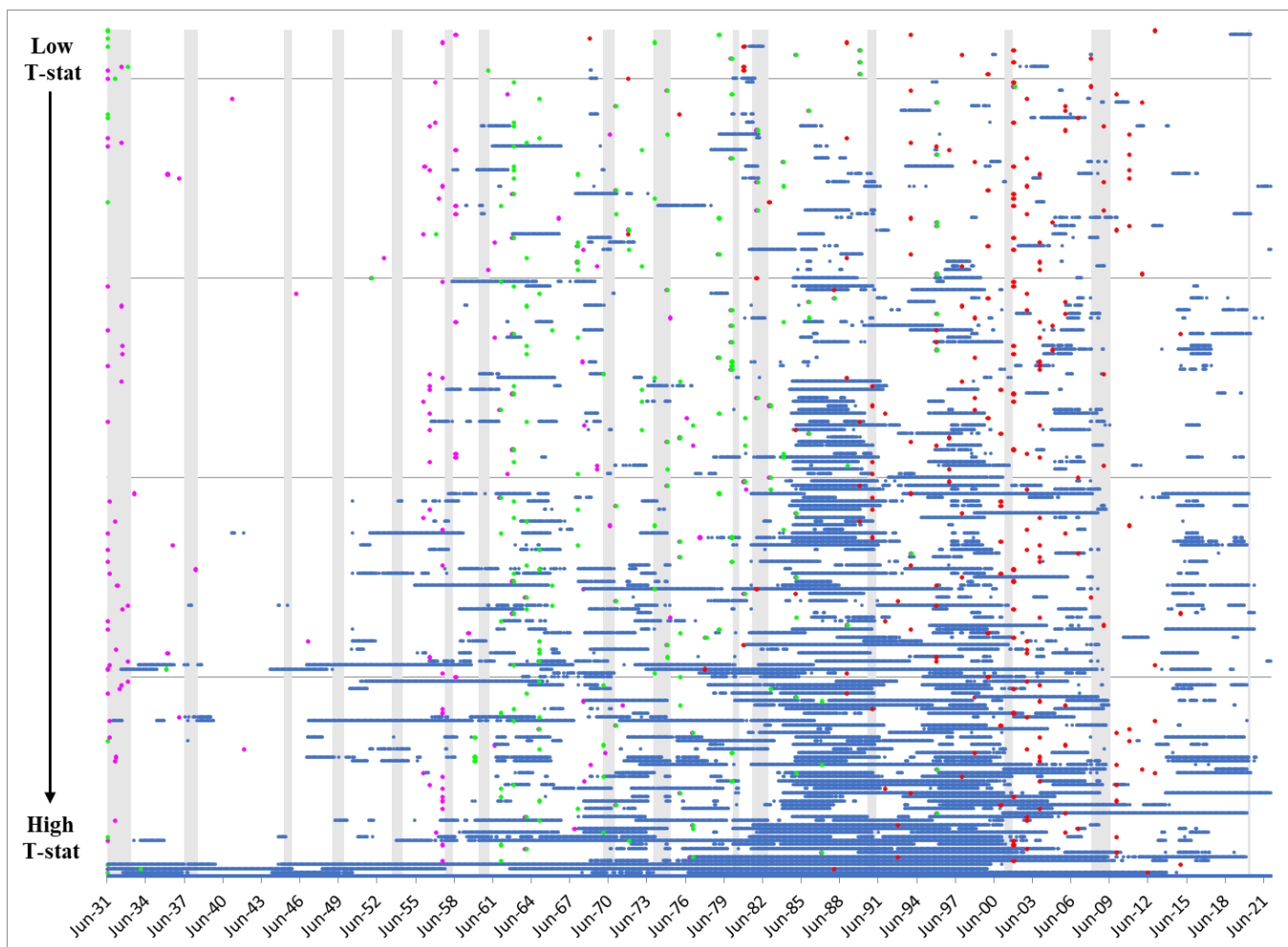


Figure 6. Simulated distributions for average significance spell lengths and counts. This figure shows simulated distributions of the cross-factor average of individual factors' average spell lengths and spell counts, when each factor has a constant return premium. We estimate for each factor a constant alpha, beta and residual volatility by means of a regression of factor returns on excess market returns. We create a simulated time series of market returns calibrated to the sample mean and standard deviation of the market over the sample period, and generate a simulated time series of returns for each of the 212 factors based on the simulated market returns, estimated factor alpha and beta, and estimated factor residual volatility, with the length of each factor's simulated time series matched to the number of sample observations for the factor return. We then estimate rolling 60-month regressions of simulated factor returns on simulated market returns and obtain both the count and average length of significance spells for each simulated factor, when significance is assessed based on t-statistics ranging from 1.96 to 4.00. Having done so, we compute the cross-factor average of the spell counts and spell lengths (corresponding to the sample data reported in Table 4). We repeat the simulation 2,000 times to obtain a distribution of the average cross-sectional factor spell lengths and counts. Panel A displays the simulated distributions for the cross-factor average of the average spell lengths, while Panel B shows corresponding cross-factor average spell counts. The red dashed lines display the corresponding sample outcomes.

Panel A: Simulated distribution of average spell length.

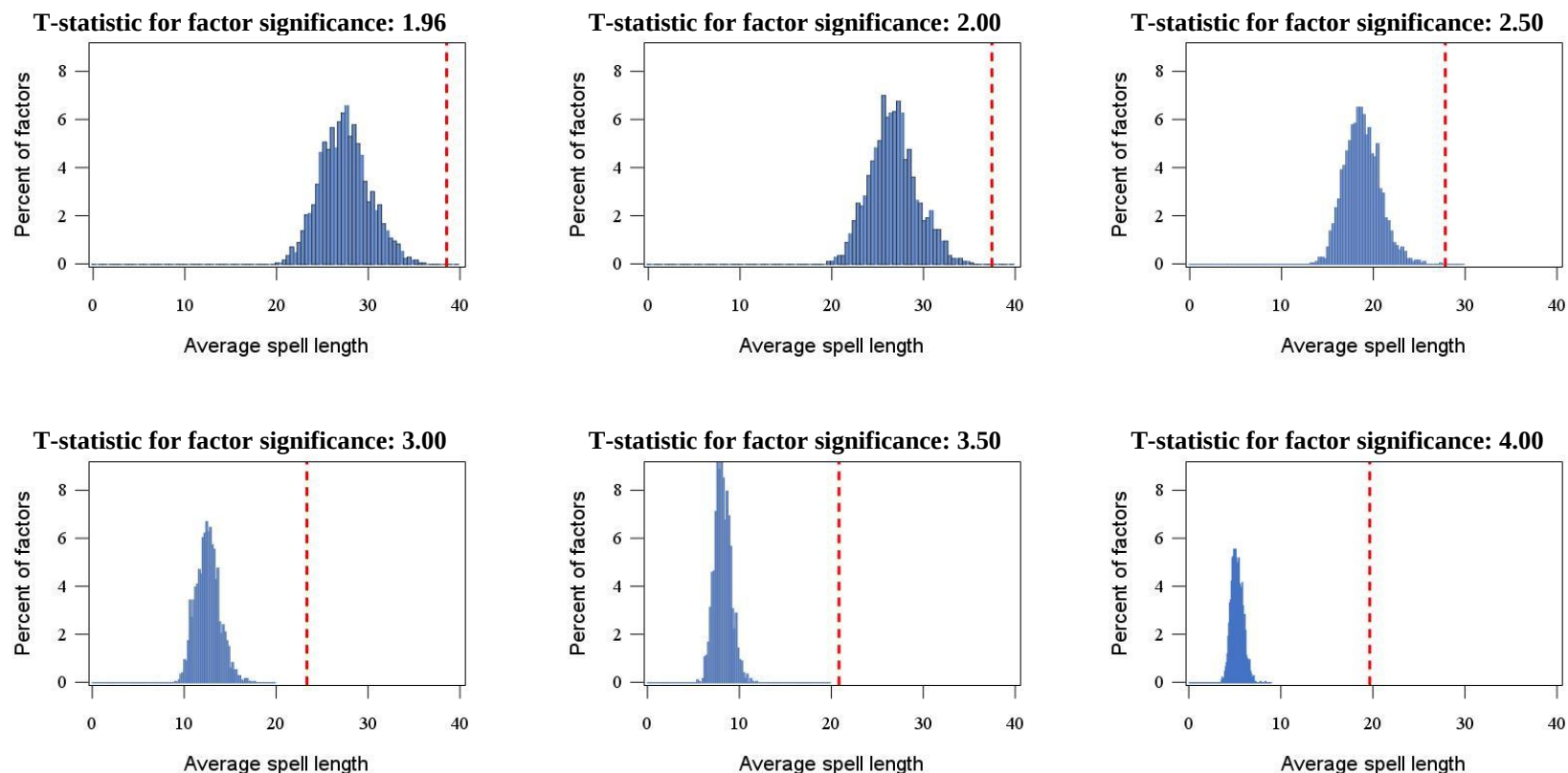
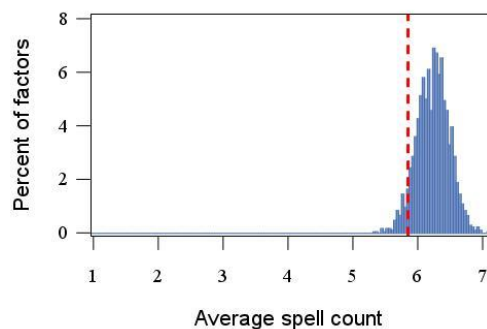


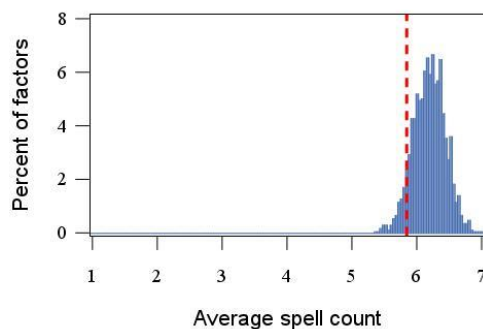
Figure 6 continued.

Panel B: Simulated distribution of average spell count.

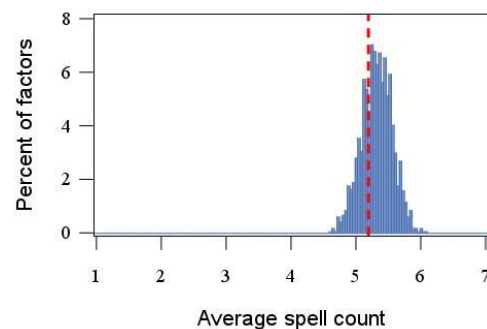
T-statistic for factor significance: 1.96



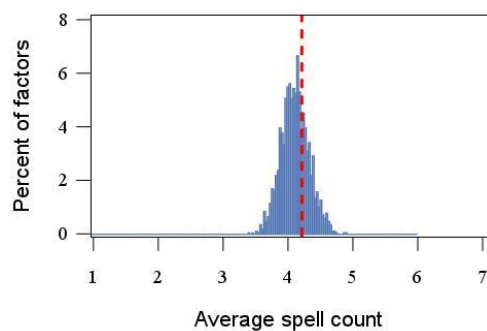
T-statistic for factor significance: 2.00



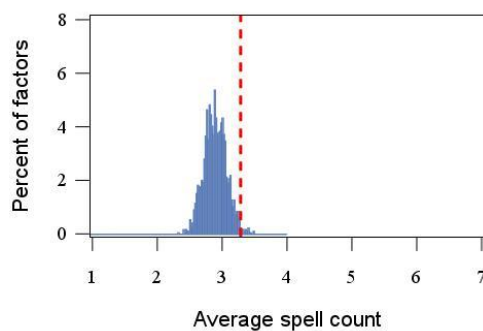
T-statistic for factor significance: 2.50



T-statistic for factor significance: 3.00



T-statistic for factor significance: 3.50



T-statistic for factor significance: 4.00

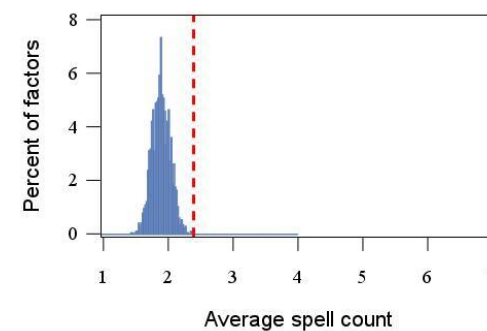


Table 1. Out-of-sample Sharpe ratios for portfolios consisting of varying numbers of principal components. This table provides descriptive statistics regarding the Sharpe ratios shown in Figure 2. We tabulate the mean and standard deviation of the time series of 5 different annualized Sharpe ratios formed from portfolios of PCs: 1) the SR from a portfolio of the first 5 PCs, 2) the average SR across portfolios formed by increasing numbers of PCs from six to the maximum number of PCs, 3) the SR from a portfolio of a predicted number of PCs, where the number of PCs is equal to the minimum of either the number of in-sample significant factors or maximum number of PCs, 4) the SR from a portfolio of the maximum number of principal components, and 5) the maximum SR attainable from any number of PCs. The “Difference from SR of 5 PCs” shows the increase in the Sharpe ratio of a given portfolio relative to the portfolio formed by the first 5 PCs. The PCs are calculated from the factor returns from $t - 59$ to t . The portfolios are formed using the out-of-sample data from $t + 1$ to $t + 36$. See Section III for the full methodology. The statistics tabulated under the heading “PCs from all factors” corresponds to the Sharpe ratios shown in Panel A of Figure 2, while those under “PCs from significant factors only” correspond to those shown in Panel B. A factor is considered significant if it has a CAPM alpha t-statistic greater than 2.0. The t-statistic on the difference of Sharpe ratios from the 5-PC portfolio is calculated using Hansen-Hodrick standard errors with a bandwidth of 60. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels.

	PCs from all factors					PCs from significant factors only			
	Average		Predicted			Average			
	5 PCs	6-Max PCs	PCs	Max PCs	Max SR	5 PCs	6-Max PCs	Max PCs	Max SR
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Mean	1.281	2.746	3.299	2.840	4.183	2.081	3.326	2.687	4.526
Std dev	0.898	1.200	1.696	1.975	1.719	1.127	1.484	1.433	1.996
N	1086	1086	1086	1086	1086	1047	1042	1085	1085
Difference from SR of 5 PCs		1.465***	2.018***	1.559***	2.902***		1.245***	0.606***	2.445***
t-statistic		7.31	6.14	3.28	8.72		4.40	2.83	5.88
% difference		114%	157%	122%	227%		60%	29%	117%

Table 2. Out-of-sample Sharpe ratios for equal-weighted portfolios consisting of varying numbers of principal components. This table provides descriptive statistics regarding the Sharpe ratios shown in Figure 3. We tabulate the mean and standard deviation of the time series of five different annualized Sharpe ratios formed from portfolios of PCs: 1) the SR from a portfolio of the first 5 PCs, 2) the average SR across portfolios formed by increasing numbers of PCs from six to the maximum number of PCs, 3) the SR from a portfolio of a predicted number of PCs, where the number of PCs is equal to the minimum of either the number of in-sample significant factors or maximum number of PCs, 4) the SR from a portfolio of the maximum number of principal components, and 5) the maximum SR attainable from any number of PCs. The “Difference from SR of 5 PCs” shows the increase in the Sharpe ratio of a given portfolio relative to the portfolio formed by the first 5 PCs. The PCs are calculated from the factor returns from $t - 59$ to t . The portfolios are formed using the out-of-sample data from $t + 1$ to $t + 36$. A factor is considered significant if it has a CAPM alpha t-statistic greater than 2.0. See Section III for the full methodology. The t-statistic on the difference of Sharpe ratios from the 5-PC portfolio is calculated based on Hansen-Hodrick standard errors with a bandwidth of 60 are in parentheses. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels.

	PCs from all factors				
	Average		Predicted		Max SR
	5 PCs	6-Max PCs	PCs	Max PCs	
	(1)	(2)	(3)	(4)	(5)
Mean	1.432	2.633	2.904	3.238	3.292
Std dev	0.615	0.990	1.297	1.246	1.274
N	1086	1086	1086	1086	1086
Difference from SR of 5 PCs		1.201***	1.472***	1.806***	1.860***
t-statistic		10.98	6.53	10.84	11.13
% difference		84%	103%	126%	130%

Table 3. Relation between the number of significant in-sample factors and out-of-sample Sharpe ratios. For each month t , we compute the factor loadings for the in-sample principal components for two sets of factor returns from $t - 59$ to t . We then use the factor loadings to form out-of-sample principal components from $t + 1$ to $t + 36$. We also compute the number of significant factors by counting the total number of factors with a CAPM alpha that has a t -statistic greater than 2.0 over the same in-sample period. The maximum out-of-sample Sharpe ratio is obtained from the optimal portfolio of out-of-sample PCs. All Sharpe ratios are annualized. Panel A shows the results of regressing the maximum out-of-sample Sharpe ratios and the number of PCs that make up the maximum out-of-sample SR portfolio on the number of significant factors. Hansen-Hodrick standard errors with a bandwidth of 60 are in parentheses. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels.

	Portfolio of principal components obtained from:			
	All factors		Significant factors	
	Max SR	Num PCs	Max SR	Num PCs
	(1)	(2)	(3)	(4)
Number of significant factors	0.01 (0.01)	0.12** (0.06)	0.02*** (0.01)	0.51*** (0.08)
Intercept	2.66*** (0.45)	46.83*** (4.31)	2.58*** (0.36)	9.75** (4.32)
R-squared	0.09	0.19	0.23	0.74
N	1086	1086	1086	1086

Table 4. Summary statistics of factor significance spells across various thresholds of significance. For each month t , we regress each factor's monthly returns from $t - 59$ to t on the market's monthly excess returns to obtain each factor's CAPM alpha. A factor is significant at month t if the t -statistic of its CAPM alpha exceeds one of the various thresholds listed in the table. Factors must have 60 non-missing returns over the alpha estimation period. A significance spell for a given factor is the number of months (i.e., spell length) the factor is continuously significant. Panel A shows the proportion of factors that exhibit at least one significance spell before (after) the sample period of the original paper to identify the factor. Panel B provides summary statistics on the number of significance spells for the cross-section of factors. Panel C computes each factor's average length of a spell and shows summary statistics of this measure for the cross-section of factors conditional on having at least one significance spell. The exceptions are that "Abs min" and "Abs max" show the absolute minimum and maximum spell length of all factors. ***, **, * represent significance at the 10%, 5% and 1% level relative to the simulated distribution in Figure 6.

Panel A: Proportion of factors with at least one significance spell

t-statistic	p-value	% significant:	
		before original sample	after original sample
1.96	0.050	78.6	92.4
2.00	0.046	78.6	92.4
2.50	0.012	68.3	82.0
3.00	0.003	55.6	66.8
3.50	0.001	43.7	50.7
4.00	0.000	23.8	34.1

Panel B: Number of significance spells per factor

t-statistic	p-value	Cross-sectional statistics of factors' spell counts				
		Mean	SD	Median	Min	Max
1.96	0.050	5.9*	4.0	5	0	18
2.00	0.046	5.8*	4.0	5	0	22
2.50	0.012	5.2	3.9	5	0	22
3.00	0.003	4.2	3.7	3	0	25
3.50	0.001	3.3**	3.6	2	0	22
4.00	0.000	2.4***	3.1	1	0	22

Panel C: Average length of significance spell

t-statistic	p-value	Cross-sectional statistics of factors' average spells						
		Mean	SD	Median	Min	Max	Abs Min	Abs Max
1.96	0.050	38.5***	48.7	25.8	1	360	1	723
2.00	0.046	37.5***	48.8	23.2	1	360	1	722
2.50	0.012	27.8***	36.4	17.8	1	360	1	669
3.00	0.003	23.3***	29.5	15.3	1	192	1	664
3.50	0.001	20.8***	25.7	12.0	1	192	1	413
4.00	0.000	19.7***	21.1	11.6	1	106	1	408

Table 5. Out-of-sample Sharpe ratios for portfolios consisting of varying numbers of principal components for alternative principal component extraction methods. This table presents the mean and standard deviation of the time series of 5 different annualized Sharpe ratios formed from portfolios of PCs: 1) the SR from a portfolio of the first 5 PCs, 2) the average SR across portfolios formed by increasing numbers of PCs from six to the maximum number of PCs, 3) the SR from a portfolio of a predicted number of PCs, where the number of PCs is equal to the minimum of either the number of in-sample significant factors or maximum number of PCs, 4) the SR from a portfolio of the maximum number of principal components, and 5) the maximum SR attainable from any number of PCs. The “Difference from SR of 5 PCs” shows the increase in the Sharpe ratio of a given portfolio relative to the portfolio formed by the first 5 PCs. The PCs are calculated from the factor returns from $t - 59$ to t . The portfolios are formed using the out-of-sample data from $t + 1$ to $t + 36$. Panel A presents results obtained using the Lettau and Pelger (2020) extraction method and Panel B shows results obtained using the BDLP (2023) extraction method. See Section III for the full methodology. Similar results using the original papers’ window length of 240 months are found in Appendix Table A2. The t-statistic on the difference of Sharpe ratios from the 5-PC portfolio is calculated using Hansen-Hodrick standard errors with a bandwidth of 60. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels.

Panel A: LP (2020) extraction method

	PCs from all factors					PCs from significant factors only			
	Average		Predicted		Max SR	Average		Max PCs	Max SR
	5 PCs	6-Max PCs	PCs	Max PCs		5 PCs	6-Max PCs		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Mean	0.897	1.296	1.469	1.471	1.754	1.018	1.424	1.596	2.012
Std dev	0.895	1.083	1.345	1.345	1.284	0.970	1.272	1.553	1.515
N	641	641	641	641	641	641	641	641	641
Difference from SR of 5 PCs		0.400***	0.572***	0.574***	0.857***		0.405**	0.577**	0.993***
t-statistic		3.50	3.19	3.21	5.18		2.12	2.04	3.92
% difference		45%	64%	64%	96%		40%	57%	98%

Panel B: BDLP (2023) extraction method

	PCs from all factors					PCs from significant factors only			
	Average		Predicted		Max SR	Average		Max PCs	Max SR
	5 PCs	6-Max PCs	PCs	Max PCs		5 PCs	6-Max PCs		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Mean	1.010	1.474	1.711	1.710	2.051	1.306	1.926	2.188	2.655
Std dev	0.973	1.203	1.515	1.515	1.474	1.157	1.516	1.846	1.902
N	641	641	641	641	641	641	641	641	641
Difference from SR of 5 PCs		0.464***	0.701***	0.700***	1.041***		0.620***	0.883***	1.350***
t-statistic		5.75	5.25	5.24	7.35		3.45	3.26	5.14
% difference		46%	69%	69%	103%		48%	68%	103%

Table 6. Out-of-sample Sharpe ratios of portfolios constructed from principal components from the set of factors formed using WRDS financial ratios (WFR). The table presents the mean and standard deviation of the time series of 5 different annualized Sharpe ratios formed from portfolios of PCs: 1) the

SR from a portfolio of the first 5 PCs, 2) the average SR across portfolios formed by increasing numbers of PCs from six to the maximum number of PCs, 3) the SR from a portfolio of a predicted number of PCs, where the number of PCs is equal to the minimum of either the number of in-sample significant factors or maximum number of PCs, 4) the SR from a portfolio of the maximum number of principal components, and 5) the maximum SR attainable from any number of PCs. The “Difference from SR of 5 PCs” shows the increase in the Sharpe ratio of a given portfolio relative to the portfolio formed by the first 5 PCs. The PCs are calculated from the factor returns from $t - 59$ to t . The portfolios are formed using the out-of-sample data from $t + 1$ to $t + 36$. A description of the factor construction from the WRDS financial ratios is in Section V.b. See Section III for the full methodology. The t-statistic on the difference of Sharpe ratios from the 5-PC portfolio is calculated using Hansen-Hodrick standard errors with a bandwidth of 60. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels.

	PCs from all factors					PCs from significant factors only			
	Average		Predicted			Average			
	5 PCs	6-Max PCs	PCs	Max PCs	Max SR	5 PCs	6-Max PCs	Max PCs	Max SR
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Mean	1.134	2.998	2.122	1.849	4.688	1.371	2.119	1.627	2.334
Std dev	1.044	1.489	2.089	1.108	2.019	1.163	1.664	1.514	1.851
N	563	563	563	563	563	410	329	533	533
Difference from SR of 5 PCs		1.864***	0.988**	0.715***	3.554***		0.748*	0.256*	0.964***
t-statistic		5.63	1.99	3.96	7.36		1.91	1.75	2.72
% difference		164%	87%	63%	313%		55%	19%	70%

Table 7. Out-of-sample Sharpe ratios of portfolios constructed from principal components from factors constructed that exclude small or less traded stocks. This table shows the annualized out-of-sample Sharpe ratios of portfolios formed from varying number of principal components based on Chen and Zimmerman (2022) factors that are formed on the subset of stocks obtained from either excluding microcaps or excluding low-volume stocks. A microcap stock is a stock with a market capitalization less than the 20th percentile of the NYSE market capitalization. A low-volume stock has an average daily turnover in the previous month less than the NYSE 20th percentile of turnover in the same period. We tabulate the mean and standard deviation of the time series of 5 different annualized Sharpe ratios formed from portfolios of PCs: 1) the SR from a portfolio of the first 5 PCs, 2) the average SR across portfolios formed by increasing numbers of PCs from six to the maximum number of PCs, 3) the SR from a portfolio of a predicted number of PCs, where the number of PCs is equal to the minimum of either the number of in-sample significant factors or maximum number of PCs, 4) the SR from a portfolio of the maximum number of principal components, and 5) the maximum SR attainable from any number of PCs. The “Difference from SR of 5 PCs” shows the increase in the Sharpe ratio of a given portfolio relative to the portfolio formed by the first 5 PCs. The PCs are calculated from the factor returns from $t - 59$ to t . The portfolios are formed using the out-of-sample data from $t + 1$ to $t + 36$. Panel A shows results obtained from the set of factors formed from the subset of stocks excluding microcaps. Panel B shows results obtained from the set of factors excluding low-turnover stocks. A factor is considered significant if it has a CAPM alpha t-statistic greater than 2.00 during the in-sample estimation window. See Section V.c for the full methodology. The t-statistic on the difference of Sharpe ratios from the 5-PC portfolio is calculated using Hansen-Hodrick standard errors with a bandwidth of 60. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels.

Panel A: Factors formed by excluding microcap stocks

	PCs from all factors					PCs from significant factors only			
	Average		Predicted			Average			
	5 PCs	6-Max PCs	PCs	Max PCs	Max SR	5 PCs	6-Max PCs	Max PCs	Max SR
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Mean	1.076	1.876	2.047	1.719	2.910	1.595	2.324	2.018	3.209
Std dev	0.802	0.831	1.138	1.028	1.014	1.010	1.061	1.149	1.255
N	1086	1086	1086	1086	1086	1056	1046	1086	1086
Difference from SR of 5 PCs		0.800***	0.970***	0.642***	1.834***		0.730***	0.423***	1.614***
t-statistic		8.64	5.44	3.38	12.68		4.67	2.89	7.70
% difference		74%	90%	60%	170%		46%	27%	101%

Panel B: Factors formed by excluding low-volume stocks

	PCs from all factors					PCs from significant factors only			
	Average		Predicted			Average			
	5 PCs	6-Max PCs	PCs	Max PCs	Max SR	5 PCs	6-Max PCs	Max PCs	Max SR
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Mean	1.384	2.519	2.778	2.339	3.731	2.035	2.929	2.366	3.927
Std dev	0.953	1.104	1.341	1.518	1.384	1.063	1.262	1.257	1.650
N	1086	1086	1086	1086	1086	1040	1027	1084	1084
Difference from SR of 5 PCs		1.135***	1.394***	0.955***	2.347***		0.893***	0.331***	1.891***
t-statistic		7.04	6.59	2.86	11.28		4.29	2.68	6.70
% difference		82%	101%	69%	170%		44%	16%	93%

Table 8. Comparison of Sharpe ratios after accounting for transaction costs. This table shows the Sharpe ratios obtained from holding the 1-month out-of-sample portfolio of PCs in the presence of transaction costs. For each month t , we compute the factor loadings for the in-sample principal components for the set of 212 Chen and Zimmerman (2022) factors from $t - 59$ to t . We construct in-sample principal components for $t - 59$ to t and out-of-sample principal components for date $t + 1$. We also compute the number of factors that have a significant (t-statistic > 2) CAPM alpha over the same window. At date $t + 1$ we form a portfolio of OOS PCs, where the number of PCs is chosen to be equal to the number of significant factors in the in-sample window. The weights applied are the mean-variance weights obtained using the same number of PCs in the in-sample window. The Sharpe ratio is computed over the full time-series of OOS returns starting in 1984. Section V.d describes the method in detail.

Model	Sharpe ratio
VW Market	0.527
Time-varying PCA - 1 month OOS	2.300
Time-varying PCA - 1 month OOS (trading costs: 10/20 bps)	1.773
Time-varying PCA - 1 month OOS (trading costs: 20/40 bps)	1.009

Table 9. Pricing performance the rolling estimation method versus canonical models. This table shows the performance of various models in pricing the 212 factors of Chen and Zimmerman (2022). For each model, we compute the SDF using the value-weighted mean-variance portfolio return. For the FF 3-factor, 5-factor, Q-factor, the mean-variance weights are computed using monthly returns from 1963 to 1983, and applied to monthly returns from 1984 to 2024 to form an out-of-sample SDF. For the 5-PC PCA model, we extract the first 5 principal components from 1963 to 2024 and then compute the mean-variance weights of these PCs from 1963 to 1983 and apply the weights to the PCs from 1984 to 2024 to form the out-of-sample SDF. For the time-varying PCA 1-month model, we compute the PC loadings and mean-variance weights using rolling 60-month windows at date $t - 59$ to t and apply these weights at $t + 1$ to form the out-of-sample SDF. We present 4 measures of pricing performance: 1) the Sharpe ratio of the out-of-sample SDF, 2) the GRS Test statistic from using the out-of-sample SDF to price the 212 factors during the out-of-sample period, 3) the Hansen-Jaganathan distance of the SDF, and 4) the mean absolute alpha computed from the individual alphas obtained for each factor relative to the SDF. Section VI describes the method in detail.

Model	Sharpe ratio	GRS Test	HJ distance	Mean abs alpha (bps)
FF 3-factor	0.684	10.312	0.123	42.9
FF 5-factor	1.584	8.233	0.130	32.9
Q-factor	1.416	8.724	0.133	28.6
PCA (5 PCs)	1.315	9.151	0.124	42.9
Time-varying PCA - 1 month OOS	2.300	4.984	0.108	19.3

Table 10. Regression of number of significant factors on macroeconomic variables and firm characteristics.

This table shows the results of regressing the number of significant factors in each period on various economic measures at each month for the sample of factors from 1968-2021. For each month t , we regress each factor's monthly returns from $t - 59$ to t on the market's monthly excess returns to obtain each factor's CAPM alpha. A factor is significant at month t if the t -statistic of its CAPM alpha exceeds 2.00. Factors must have 60 non-missing returns over the alpha estimation period. The number of public firms is a count of all common stocks at t traded on the NYSE, Nasdaq or Amex at month t . The mean institutional ownership is the fraction of a firm's shares outstanding held by 13-f firms. The NBER recession indicator is an indicator equal to one if the month is classified as an NBER recession and zero otherwise. The unemployment rate is the number of unemployed as a percentage of the labor force as provided by the U.S. bureau of labor statistics. The 90-day T-bill rate is the 3-month Treasury Bill Secondary Market Rate and the 10-year treasury note yield spread is the difference of the market yield on U.S. treasury securities at a 10-year constant maturity and the 90-day T-bill rate. The percentage of dividend-paying firms is the total number of common stocks which have paid a dividend in the previous 12 months divided by the number of firms at month t . The economic complexity index is a measure of economic complexity constructed by Simoes and Hidalgo (2011). Diversity of firm characteristics is a measure of diversity in the cross-sectional characteristics across firms. To facilitate interpretation, we standardize each of the following variables relative to its own time series standard deviation. See Appendix Table A1 for a complete description of the measures. Hansen-Hodrick standard errors (using a bandwidth of 60) are in parentheses. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels.

	Dep var: Number of significant factors			
	(1)	(2)	(3)	(4)
Number of public firms	10.47** (4.54)	31.17*** (8.91)		
Economic complexity index	4.16** (1.87)		12.19*** (3.85)	
Diversity of firm characteristics	8.51** (3.55)			16.29*** (5.81)
Unemployment rate	-12.37*** (1.85)			
90-day T-Bill yield	15.55*** (2.48)			
10-year T-Note yield spread	11.96*** (1.72)			
% dividend-paying firms	-5.09 (4.07)			
Mean Amihud illiquidity	4.88 (3.56)			
NBER recession indicator	-5.33** (2.22)			
Adjusted R-squared	0.79	0.51	0.21	0.29
N	612	648	648	612

Appendix: Additional Figures and Tables

Figure A1. In-sample and out-of-sample SRs by individual principal component. This figure shows the time series average of the rolling in-sample (IS) and out-of-sample (OOS) Sharpe ratios as well as the slope coefficient from a regression of the OOS SR on the IS SR. For each month t , we compute the in-sample principal components for the full sample of factor returns from $t - 59$ to t . We also use the in-sample factor loadings to form out-of-sample principal component factors from $t + 1$ to $t + 36$. We then calculate the in-sample SR for each PC individually and the corresponding out-of-sample SR for the same PC. Table IA-2 in the internet appendix show summary statistics of all SRs in the time series by principal component number.

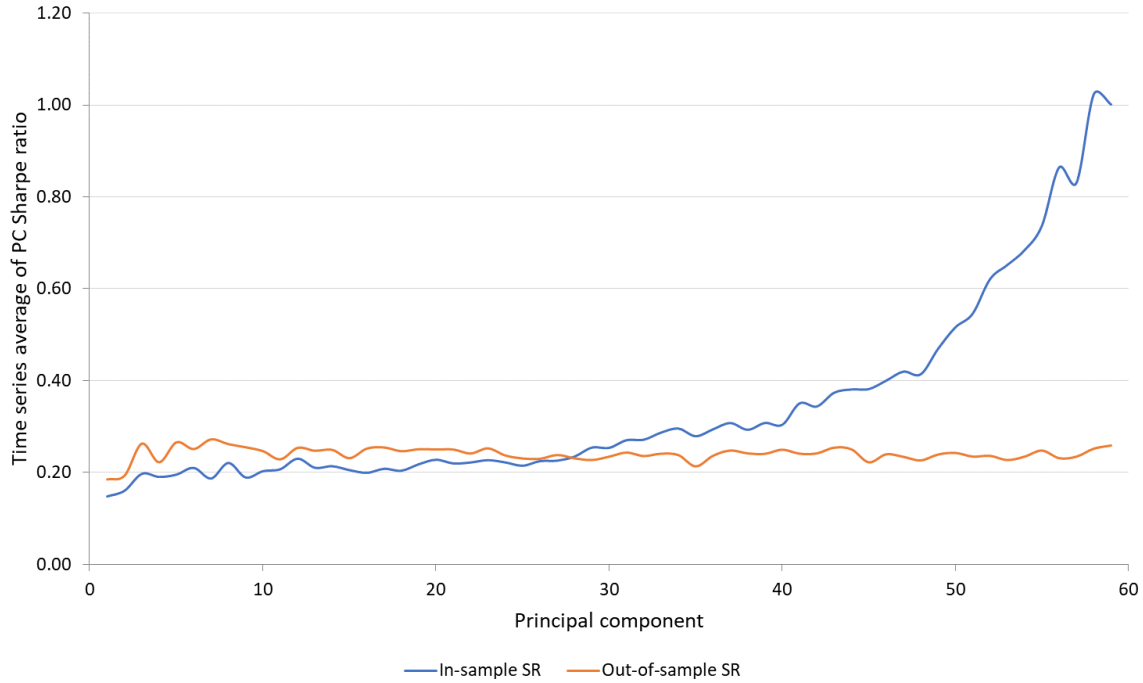


Figure A2. Trade-off between number of characteristics employed and the number of stock-month observations that can be used for KPS (2019) IPCA estimation. This table shows the inverse relationship between the number of underlying characteristics employed in the IPCA method of Kelly, Pruitt and Su (2019) and the stock-month observations available for that analysis. The tradeoff arises because at each date t , stocks can be included in the analysis only if they have all non-missing observations for all employed characteristics. Thus, employing more characteristics reduces both the length of the time series and the number of stocks that can be employed.

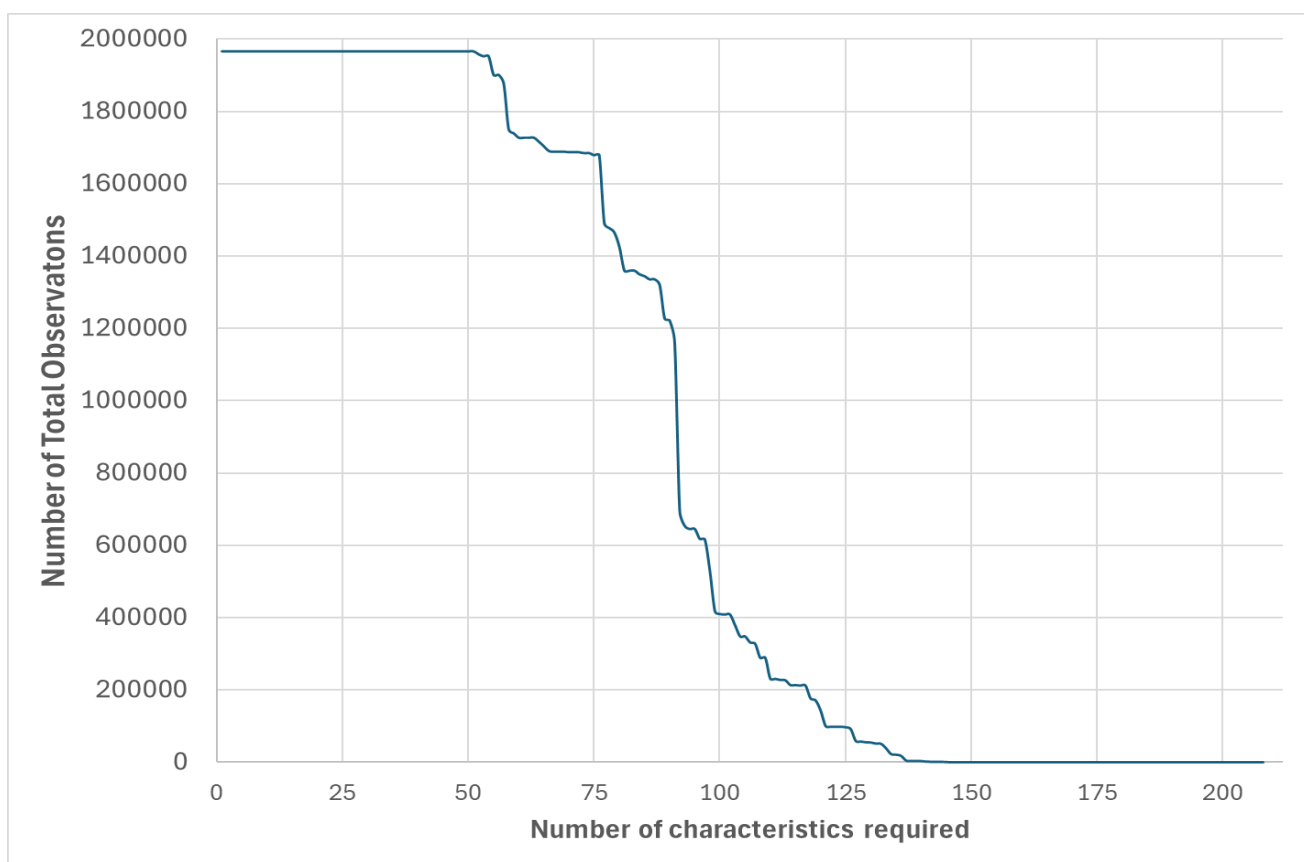


Table A1. Variable definitions This table summarizes the various variables we use throughout the analysis. The variables are listed in order of appearance in the paper.

<u>Variable name</u>	<u>Description</u>
Number of significant factors	The total number of significant factors at each month t . At each month t , we regress each factor's returns from $t-59$ to t on the market's excess returns over the same period to obtain the factor's CAPM alpha. A factor is considered significant if the t -statistic of its CAPM alpha is greater than 3.00. To be included, the factor must have zero non-missing returns over the 60-month period. Factors come from Chen and Zimmerman (2022) and are categorized as "clear" or "likely" predictors.
Standard deviation of stocks' alphas	The equal-weighted (value-weighted) cross-sectional standard deviation of stocks' alphas at month t . At each month t , we regress each factor's returns from $t-59$ to t on the market's excess returns to obtain the stock's CAPM alpha. To be included, the stock must have zero non-missing returns over the 60-month period. Stocks are all common stocks (CRSP share codes 10 or 11) listed on the NYSE, AMEX and NASDAQ. The value-weighted cross-sectional standard deviation is weighted by each stock's market capitalization at $t-61$. Units are expressed as percentage points. This measure is standardized across the sample for ease of interpretability.
NBER recession	A recession indicator equal to 1 if the economy at month t was in a recession as defined by the National Bureau of Economic Research. Data can be obtained here: https://fred.stlouisfed.org/series/USREC
25 Size and Book-to-Market portfolios	Monthly equal-weighted returns from 25 size and book-to-market portfolios provided on Ken French's website: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
30 Fama-French industry portfolios	Monthly equal-weighted returns from Fama-French 30 industry portfolios provided on Ken French's website: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
Significance spell of factor	The number of consecutive months for which a factor remains significant.
Mean standard error of alphas	The equal-weighted average standard error of CAPM alphas for all stocks at month t . The value-weighted mean standard error is weighted by each stock's market capitalization at $t-60$. This measure is standardized across the sample for ease of interpretability.
Mean residual volatility	The average residual volatility across stocks obtained from the 60-month stock-level CAPM regressions at each month t . This measure is standardized across the sample for ease of interpretability.
Number of public firms	The total number of CRSP common stocks listed on the NYSE, AMEX or NASDAQ at time t . This measure is standardized across the sample for ease of interpretability.
Number of public firms at beginning of estimation period	The total number of CRSP common stocks listed on the NYSE, AMEX or NASDAQ at time $t-60$. This measure is standardized across the sample for ease of interpretability.
Number of public firms with alpha at t	The total number of CRSP common stocks at month t listed on the NYSE, AMEX or NASDAQ for which an alpha can be calculated (i.e., stock has zero non-missing returns from $t-59$ to t). This measure is standardized across the sample for ease of interpretability.
Mean absolute alpha	The equal-weighted average of the absolute value of alpha for all factors at month t . This measure is standardized across the sample for ease of interpretability.

Mean standard error of alphas	The equal-weighted average standard error of alphas for all factors at month t . This measure is standardized across the sample for ease of interpretability.
Unemployment rate	The percentage of the labor force unemployed at t as determined by the US Bureau of Labor Statistics. This measure is standardized across the sample for ease of interpretability. Data can be obtained here: https://fred.stlouisfed.org/series/UNRATE
Fed funds rate	The federal funds rate at the end of each month t . This measure is standardized across the sample for ease of interpretability. Data can be obtained here: https://fred.stlouisfed.org/series/FEDFUNDS
10-year Treasury Bond yield	The 10-year Treasury bond yield at the end of each month t . This measure is standardized across the sample for ease of interpretability. Data can be obtained here: https://fred.stlouisfed.org/series/DGS10
% of dividend-paying firms	The total number of common stocks which pay a dividend divided by the total number of common stocks at each month t . A stock is defined as paying a dividend if at least one dividend was paid over the previous year. This measure is standardized across the sample for ease of interpretability.
Mean institutional ownership	The average institutional ownership across stocks at month t . For each stock at month t , the percentage of institutional ownership is determined by the total number of shares held by institutions divided by the total number of shares outstanding. Institutional shareholdings are obtained from Thomson-Reuters 13-F database. This measure is standardized across the sample for ease of interpretability.
Economic complexity index	An annualized measure of economic complexity based on the complexity of trade activities within the United States. Each month t uses the measure from December of the most previous year. This measure is standardized across the sample for ease of interpretability. Data can be obtained here: https://oec.world/en/rankings/legacy/eci
Mean Amihud Illiquidity	For each stock in each month, we compute the Amihud (2002) illiquidity measure using daily data. We require at least 10 trading days in a month. We then average this measure across all stocks in that month.
Diversity of firm characteristics	We compute the standard deviation for each of the 212 cross-sectional characteristics across firms in each month t . We then standardize each of these characteristics' standard deviations based on the entire time series for that characteristic. We then sum all the standardized characteristics available at each month. Finally, we use the month $t+36$ measure as a measure of diversity in month t . This accounts for the fact that in the first few years a firm enters the sample, less than twenty characteristics can be computed. Thirty-six months after listing, approximately one hundred characteristics can be computed. This rapid growth reflects that many characteristics require prior accounting statement data (which is often sparsely collected at the beginning of a firm's public life), as well as prior return history. However, the fact that a given characteristic cannot yet be computed by an econometrician need not imply that market participants are unaware of the characteristic. The final measure used in the regression is standardized across the sample for ease of interpretability.

Table A2. Out-of-sample Sharpe ratios for portfolios consisting of varying numbers of principal components for alternative principal component extraction methods. This table presents the mean and standard deviation of the time series of five different annualized Sharpe ratios formed from portfolios of PCs: 1) the SR from a portfolio of the first 5 PCs, 2) the average SR across portfolios formed by increasing numbers of PCs from six to the maximum number of PCs, 3) the SR from a portfolio of a predicted number of PCs, where the number of PCs is equal to the minimum of either the number of in-sample significant factors or maximum number of PCs, 4) the SR from a portfolio of the maximum number of principal components, and 5) the maximum SR attainable from any number of PCs. The “Difference from SR of 5 PCs” shows the increase in the Sharpe ratio of a given portfolio relative to the portfolio formed by the first 5 PCs. The PCs are calculated from the factor returns from $t - 239$ to t , which follows the in-sample window lengths of the original papers. The portfolios are formed using the out-of-sample data from $t + 1$ to $t + 36$. Panel A presents results obtained using the Lettau and Pelger (2020) extraction method and Panel B shows results obtained using the BDLP (2023) extraction method. We use both the full set of factors and the subset of significant factors only. A factor is considered significant if it has a CAPM alpha t-statistic greater than 2.0. See Section V.a for the full methodology. The t-statistic on the difference of SRs from the 5-PC portfolio is calculated using Hansen-Hodrick standard errors with a bandwidth of 240. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels.

Panel A: LP (2020) extraction method

	PCs from all factors					PCs from significant factors only			
	Average		Predicted			Average			
	5 PCs	6-Max PCs	PCs	Max PCs	Max SR	5 PCs	6-Max PCs	Max PCs	Max SR
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Mean	1.341	2.286	3.265	2.528	3.230	1.355	2.444	2.524	3.471
Std dev	0.754	1.063	1.371	1.275	1.440	0.930	0.961	1.086	1.265
N	461	461	449	461	461	449	449	449	449
Difference from SR of 5 PCs		0.945***	1.924***	1.188***	1.890***		1.089***	1.169***	2.116***
t-statistic		4.22	3.74	4.67	6.54		5.18	3.62	8.77
% difference		70%	144%	89%	141%		80%	86%	156%

Panel B: BDLP (2023) extraction method

	PCs from all factors					PCs from significant factors only			
	Average		Predicted			Average			
	5 PCs	6-Max PCs	PCs	Max PCs	Max SR	5 PCs	6-Max PCs	Max PCs	Max SR
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Mean	1.410	2.814	2.950	2.950	3.835	1.584	3.331	3.265	4.483
Std dev	0.814	1.445	1.521	1.521	1.839	0.917	1.334	1.371	1.685
N	461	461	461	461	461	449	449	449	449
Difference from SR of 5 PCs		1.404***	1.540***	1.540***	2.425***		1.748***	1.681***	2.900***
t-statistic		3.94	4.58	4.58	5.06		3.82	3.37	5.13
% difference		100%	109%	109%	172%		110%	106%	183%

Table A3. Time variation in Sharpe ratios factors accounting for conditional betas following Kelly, Pruitt and Su (2019). This table provides descriptive statistics regarding the Sharpe ratios obtained from out-of-sample Sharpe ratios of portfolios formed from varying number of latent factors extracted using the Kelly, Pruitt and Su (2019) method from the 212 Chen and Zimmerman (2022) factors. We tabulate the mean and standard deviation of the time series of 5 different annualized Sharpe ratios formed from portfolios of PCs: 1) the SR from a portfolio of the first 5 PCs, 2) the average SR across portfolios formed by increasing numbers of PCs from six to the maximum number of PCs, 3) the SR from a portfolio of a predicted number of PCs, where the number of PCs is equal to the minimum of either the number of in-sample significant factors or maximum number of PCs, 4) the SR from a portfolio of the maximum number of principal components, and 5) the maximum SR attainable from any number of PCs. The “Difference from SR of 5 PCs” shows the increase in the Sharpe ratio of a given portfolio relative to the portfolio formed by the first 5 PCs. The PCs are calculated from the factor returns from $t - 59$ to t . The portfolios are formed using the out-of-sample data from $t + 1$ to $t + 36$. We results obtained from 75 latent factors extracted using the KPS (2019) methodology. We present results for the set of all latent factors and significant latent factors only, where a latent factor is considered significant if it has a CAPM alpha t-statistic > 2 during the in-sample estimation window. See Section V.e for the full methodology. The t-statistic on the difference of SRs from the 5-PC portfolio is calculated using Hansen-Hodrick standard errors with a bandwidth of 60 are in parentheses. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels.

	PCs from all factors					PCs from sig factors only			
	Average		Predicted			Average			
	5 PCs	6-Max PCs	PCs	Max PCs	Max SR	5 PCs	6-Max PCs	Max PCs	Max SR
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(9)	(10)
Mean	1.051	2.158	1.610	1.936	3.611	1.937	2.383	2.513	2.888
Std dev	0.787	0.770	0.987	1.132	1.159	0.970	1.001	1.214	1.198
N	535	535	535	535	535	517	518	535	535
Difference from SR of 5 PCs		1.108***	0.559***	0.885**	2.560***		0.447**	0.577**	0.951***
		4.09	5.51	2.12	5.21		2.26	2.27	3.85
% difference		105%	53%	84%	244%		23%	30%	49%

Table A4. Out-of-sample Sharpe ratios for portfolios consisting of varying numbers of principal components for differing estimation

windows. This table provides descriptive statistics regarding the Sharpe ratios of portfolios formed from varying number of principal components. For each month t , we compute the factor loadings for the in-sample principal components for two sets of factor returns from $t - j$ to t . We then use the factor loadings to form out-of-sample principal components from $t + 1$ to $t + k$. We tabulate the mean and standard deviation of the time series of 5 different annualized Sharpe ratios formed from portfolios of PCs: 1) the SR from a portfolio of the first 5 PCs, 2) the average SR across portfolios formed by increasing numbers of PCs from six to the maximum number of PCs, 3) the SR from a portfolio of a predicted number of PCs, where the number of PCs is equal to the minimum of either the number of in-sample significant factors or maximum number of PCs, 4) the SR from a portfolio of the maximum number of principal components, and 5) the maximum SR attainable from any number of PCs. The first row is the baseline result shown in Table 1.

Window (months)		Portfolio SR					PCs from sig factors only			
In-sample	Out-of-sample	Average		Predicted				Average		
		5 PCs	6-59 PCs	PCs	Max PCs	Max SR	5 PCs	6-59 PCs	Max PCs	Max SR
60	36	1.28	2.75	3.30	2.84	4.18	2.08	3.33	2.69	4.53
60	60	1.27	2.66	3.12	2.70	3.85	2.00	3.25	2.66	4.25
36	36	1.09	2.19	2.50	2.50	3.27	1.93	2.93	2.66	4.14
120	120	1.54	3.29	3.60	2.43	4.43	2.03	3.54	2.65	4.51
120	60	1.60	3.43	3.76	2.68	4.79	2.12	3.66	2.73	4.93
120	36	1.64	3.52	3.39	2.78	5.13	2.16	3.85	2.82	5.30